Carbon Pricing and Corporate Biodiversity Footprints: Co-Benefits and Trade-Offs

PEJMAN ABEDIFAR, SOROUSH KAMYAB, and STEVEN ONGENA *

This draft: October 2026

ABSTRACT

This paper examines how carbon pricing affects corporate biodiversity footprints (CBF) using firm-level panel data from 2018–2024. Exploiting Washington State's Cap-and-Invest Program as a quasi-experimental shock, we combine difference-in-differences estimation with propensity score matching. We find that carbon pricing initially reduces firms' overall biodiversity footprints, although the effect weakens over time. Air and water pollution-related pressures decline persistently, whereas land use and resource use pressures reverse their initial reductions and increase in later years. Rising resource use pressures may reflect greater demand for material-intensive technologies associated with the low-carbon transition, while increasing land use pressures may result from firms relocating production to less regulated jurisdictions. Heterogeneity analyses reveal stronger environmental improvements among firms with higher initial GHG-related biodiversity footprints and greater capital investment intensity, but weaker GHG-related improvements among rapidly growing firms. Overall, carbon pricing generates biodiversity co-benefits while potentially creating trade-offs in land and resource use.

Keywords: *Biodiversity; Climate Change; Corporate Biodiversity Footprint; Carbon Pricing*

JEL Classification: G34; G38; Q56; Q58

* Pejman Abedifar (Corresponding author) is at the Tehran Institute for Advanced Studies, Khatam University. Soroush Kamyab is at Bayes Business School, City St George's, University of London. Steven Ongena is at the University of Zurich, Swiss Finance Institute, KU Leuven, NTNU Business School, and CEPR. We sincerely appreciate Professor Alexander Wagner at the University of Zurich for his helpful comments. We also thank Professor Alexander Wagner and Professor Thomas Rauter at the University of Zurich for their assistance in facilitating and arranging access to the data. Email addresses: p.abedifar@teias.institute (P. Abedifar); soroush.kamyab@citystgeorges.ac.uk (S. Kamyab); steven.ongena@df.uzh.ch (S. Ongena).

I. Introduction

Carbon pricing regulations are primarily designed to reduce greenhouse gas (GHG) emissions, but their consequences for biodiversity remain largely unexplored. By raising the cost of carbon-intensive activities, these policies may encourage cleaner production and reduce environmental pressures beyond GHG emissions. At the same time, the transition toward low-carbon technologies and changes in firms' production networks may increase pressures associated with natural resource extraction and land use. This raises an important question: *Does carbon pricing reduce firms' overall biodiversity footprints, or does it shift environmental pressures across biodiversity dimensions?* We address this question by studying Washington State's Cap-and-Invest Program, enacted under the Climate Commitment Act (CCA) in May 2021, as a quasi-experimental policy shock (Bohnet et al., 2026).

Environmental regulations influence firms' production choices and environmental externalities (OECD, 2011; IPCC, 2014). While carbon pricing policies focus on GHG emissions, biodiversity-oriented regulations seek to address a wider range of ecological harms, including land use change, water pollution, and ecosystem degradation. The distinction matters because reductions in carbon emissions do not necessarily imply proportional improvements in other environmental outcomes. The substantial decline in biodiversity over recent decades (WWF, 2024) and its implications for ecosystems, economic activity, and financial stability (World Bank, 2020; IPBES, 2020; Garel et al., 2024) have increased attention to biodiversity conservation. International initiatives, including the Paris Agreement (Agreement, 2015), the Kunming Declaration (Kunming Declaration, 2021), and the Kunming–Montreal Global Biodiversity Framework (Montreal Agreement, 2022), underscore the importance of understanding whether climate-focused regulations complement or conflict with biodiversity objectives.

Carbon pricing may affect corporate biodiversity footprints through several economic and technological mechanisms. On the one hand, higher emissions costs may encourage firms to improve energy efficiency, reduce fossil fuel consumption, adopt cleaner technologies, and optimize their production processes (Martin et al., 2014; Calel and Dechezleprêtre, 2016). Because carbon-intensive activities are often associated with other harmful environmental externalities (Greenstone, 2003; Shapiro and Walker, 2018), these adjustments may generate benefits extending beyond GHG emissions. For example, lower fossil fuel combustion may also reduce sulfur dioxide, nitrogen oxides, particulate matter, and toxic emissions contributing to air and water pollution (Fowlie, 2010). Improvements in production efficiency may further reduce waste, material use, and environmental disturbance associated with extraction, transportation, and industrial processing (Berman and Bui, 2001; Copeland and Taylor, 2004). Carbon pricing may also encourage firms to modernize production systems, redesign supply chains, improve environmental monitoring, and accelerate green innovation (Porter and Linde, 1995; Calel and Dechezleprêtre, 2016). Scrutiny from financial markets and other stakeholders may reinforce these incentives (Matsumura et al., 2014; Krüger, 2015; Flammer, 2021; Bolton and Kacperczyk, 2021). Together, these channels suggest that carbon pricing can generate biodiversity co-benefits beyond its direct emissions-reduction objective.

On the other hand, adjustments toward low-carbon production may create new biodiversity pressures through input substitution and supply-chain reconfiguration. Renewable energy technologies, electrification systems, battery storage, and low-carbon infrastructure rely on substantial quantities of critical minerals and other natural resources, including lithium, cobalt, nickel, copper, and rare earth elements (IEA, 2021; Press et al., 2023). More broadly, Vanden Eynde and Vargas (2025) review how the green transition can increase demand for critical minerals and create new pressures associated with natural resource extraction. Because corporate biodiversity footprints capture both direct and indirect environmental impacts, these adjustments may increase firms' resource use-related pressures through their supply chains, even when the underlying resource extraction occurs outside the regulated jurisdiction. Similarly, firms may reorganize production networks or relocate activities to jurisdictions with less stringent environmental regulations, potentially increasing land use-related pressures.[1] Nevertheless, carbon pricing may also encourage greater material efficiency and reduced resource consumption. Consequently, its effects on resource use and land use remain theoretically ambiguous and require empirical investigation.

These mechanisms may also operate over different time horizons. Changes in fuel consumption, output, and operating efficiency can occur relatively quickly following a policy announcement (Martin et al., 2014; Dechezleprêtre et al., 2022). In contrast, technological substitution, investment in new infrastructure, and supply-chain restructuring typically require longer planning and adjustment periods (Fischer and Newell, 2008; Acemoglu et al., 2012; IEA, 2021). Consistent with this distinction, our review of firms' 10-K and 10-Q filings indicates that some companies began evaluating compliance strategies shortly after the CCA was enacted. These strategies included investments in energy efficiency, greater reliance on renewable energy, purchases of emission allowances and offsets, and adjustments to operating practices.[2] Because some adjustments may improve biodiversity outcomes while others may intensify particular pressures, both the overall effects and their evolution over time are empirical questions.

Washington's Cap-and-Invest Program provides a useful setting to examine these competing mechanisms. The CCA establishes a declining emissions cap and requires covered entities emitting at least 25,000 metric tons of CO_2 equivalent annually to obtain emission allowances. The program covers activities in energy, transportation, manufacturing, and natural gas distribution, among other sectors, and operates through the Compliance Instrument Tracking System Service (CITSS).[3] Although the Cap-and-Invest Program became operational in Jan-

[1] The relocation of carbon-intensive activities to jurisdictions with less stringent environmental regulation is commonly discussed in the carbon leakage literature (Cole, 2004; Copeland and Taylor, 2004; Levinson and Taylor, 2008; Dechezleprêtre and Sato, 2017; Dechezleprêtre et al., 2022).

[2] For example, in its Form 10-Q filed in November 2021, *Avista Corporation* stated: "*The Climate Commitment Act signed into law in May 2021 requires natural gas distribution companies to reduce overall GHG emissions 45 percent below 1990 levels by 2030, 70 percent below 1990 levels by 2040 and 95 percent below 1990 levels by 2050, which may be achieved through increased energy efficiency and conservation measures, purchased emission allowances and offsets, and purchase of renewable natural gas. The Company has begun reviewing compliance options and expects the compliance costs for these legislated actions.*"

[3] CITSS is the administrative platform used to register market participants and track allowance holdings. Official registrant reports identify covered entities and sources and distinguish their regulatory status.

uary 2023, we use the enactment of the CCA in 2021 as the policy shock to capture firms' potential anticipatory responses. The emissions threshold and broader policy design were not determined by firms' biodiversity performance. This reduces concerns that the policy was introduced or targeted based on firms' prior biodiversity outcomes, supporting its interpretation as a plausibly exogenous policy shock to corporate biodiversity footprints. To identify exposed firms, we use official CITSS registrant data on Covered Entities and Covered Sources subject to mandatory compliance, supplemented by publicly listed firms headquartered in Washington State that may be indirectly affected by the regulation.

We combine firm-level corporate biodiversity footprint (CBF) data from Iceberg Data Lab (IDL) with financial information from Compustat for the period 2018–2024. Using a difference-in-differences (DiD) framework and propensity score matching, we compare the environmental outcomes of firms exposed to the Washington policy with those of comparable control firms. The CBF measure enables us to examine not only firms' overall biodiversity pressures but also their underlying GHG, land use, resource use, air pollution, and water pollution components. Our central contribution is to assess whether a regulation targeting carbon emissions generates broader biodiversity benefits or creates trade-offs across environmental pressures, and how these responses vary across firms.

We measure firms' biodiversity pressures using the CBF framework developed by IDL, which captures annual direct and indirect environmental impacts associated with firms' operations and value chains. In addition to total CBF, the framework provides separate measures for GHG emissions, land use, resource use, air pollution, water pollution, and invasive alien species. The latter component is excluded from the empirical analysis because of limited coverage. We merge these data with Compustat financial characteristics, including firm size, leverage, profitability, liquidity, investment intensity, and market valuation. The resulting unbalanced panel comprises 2,415 U.S. firms and 12,292 firm-year observations prior to performing the propensity score matching. To improve comparability, we employ nearest-neighbor propensity score matching on financial characteristics and pre-policy GHG-related biodiversity exposure, restricting matches to firms within the same two-digit GICS industry. Under the baseline 5:1 matching specification with a caliper of 0.2, the estimation sample contains 232 firms, including 44 treated firms. Our DiD specification then compares changes in treated firms' biodiversity footprints with those of matched controls around the 2021 enactment, including firm and year fixed effects.

Our baseline results suggest that carbon pricing has contrasting effects across biodiversity dimensions. Total CBF declines significantly in 2022 and 2023, with an aggregate post-treatment estimate of approximately -0.21 log points, but the effect is no longer statistically significant in 2024. The initial reductions are accompanied by improvements in GHG-related biodiversity footprints and several non-GHG pressures. However, the dynamics differ substantially across components. Air and water pollution-related footprints remain lower throughout the post-treatment period, whereas land use and resource use show an initial decline followed by a reversal. Land use-related pressures increase significantly in 2024, while resource use-related pressures rise in 2023 and increase sharply in 2024. Nevertheless,

resource use accounts for only approximately 1.6% of total CBF among U.S. firms, suggesting that its substantial relative increase may have a more limited effect on overall biodiversity footprints. These findings indicate that carbon pricing may generate environmental co-benefits while being associated with potential trade-offs in particular biodiversity dimensions. Changes in firms' investments, supply chains, and geographical operations offer possible explanations, although our estimates do not directly establish these mechanisms.

The heterogeneity analysis further indicates that firms' responses differ according to their initial environmental exposure and financial characteristics. Firms with higher pre-policy GHG-related biodiversity footprints exhibit larger reductions in the GHG component, consistent with stronger incentives and greater opportunities for abatement. Firms with higher capital investment intensity also show larger reductions in total and GHG-related CBF, suggesting a possible role for technological upgrading. In contrast, faster-growing firms experience weaker GHG-related improvements, consistent with expansion-related scale effects that may offset efficiency gains. Larger firms exhibit weaker reductions in some non-GHG components, plausibly because their more diversified operations generate environmental pressures from multiple sources, limiting the extent to which carbon-focused adjustments translate into broader environmental improvements.

This research makes several contributions to the environmental economics and finance literature. First, whereas existing studies primarily examine the effects of carbon pricing on emissions (Greenstone, 2003; Auffhammer and Kellogg, 2011; Bushnell and Wolfram, 2012; Bu and Shi, 2021; Li et al., 2021), human well-being, and firm performance (Becker and Henderson, 2000; Copeland and Taylor, 2004; Bushnell and Wolfram, 2012; Curtis, 2018), we examine how these regulations affect corporate biodiversity footprints beyond their direct carbon-reduction objectives. Second, we contribute to the growing literature on the biodiversity consequences of environmental regulation. For instance, Liang et al. (2020) document improvements in bird populations following U.S. air pollution regulation, while Zhou and Almond (2026) examine habitat-quality effects of voluntary carbon offset projects using satellite-based measures. We complement these studies by focusing on the firms and examining both direct and indirect biodiversity pressures associated with its operations and value chains. The decomposition of CBF into individual pressure categories allows us to identify how environmental improvements in some dimensions may coexist with increases in others. Our analysis also relates to the literature on the broader consequences of the green transition for natural resources. For example, Vanden Eynde and Vargas (2025) review how increasing demand for critical minerals may generate environmental and socioeconomic pressures in resource-producing regions. We complement this broader perspective by examining the firm-level biodiversity footprint responses to a specific carbon pricing policy, without directly testing changes in mineral extraction or conflict. Third, whereas Garel et al. (2024) investigate the relationship between firm characteristics and biodiversity footprints, we examine how those characteristics shape firms' responses to carbon pricing, identifying potential differences in environmental adjustment across firms.

Our findings also have implications for environmental policy design. Washington State's

declining emissions cap, allowance auctions, and use of auction revenues for climate-related investments illustrate how carbon pricing can be combined with broader climate policy objectives. Nevertheless, our results suggest that a policy directed at GHG emissions may not fully address environmental pressures associated with the transition toward lower-carbon production, especially those involving land and natural resources. Complementary measures encouraging energy and material efficiency, cleaner production technologies, recycling, circular production processes, and lower material-intensity investments may help limit these pressures (Acemoglu et al., 2012). In addition, the resource-governance literature emphasises the importance of transparency and regulation in extractive supply chains (Vanden Eynde and Vargas, 2025). Greater attention to the sourcing of material-intensive inputs could therefore complement carbon pricing, although the effectiveness of these measures for biodiversity outcomes requires separate evaluation. Such measures could improve the alignment between decarbonization and biodiversity conservation without diminishing the emissions-reduction incentives created by carbon pricing.

The remainder of the paper is organized as follows. Section II describes the institutional background and study setting. Section III presents the data sources and matching procedure. Section IV outlines the identification strategy and empirical framework. Section V reports the baseline and heterogeneous treatment effects. Section VI presents placebo tests and alternative matching specifications. Section VII concludes.

II. Study Setting

We use Washington State's Cap-and-Invest Program as the setting for this study. Enacted as part of the Climate Commitment Act (CCA) on May 17, 2021, this program represents one of the most ambitious market-based approaches to reducing greenhouse gas emissions in the United States. The program aims to help Washington achieve its goal of reducing emissions to 45 percent below 1990 levels by 2030 and reaching net-zero emissions by 2050.

The regulation is modeled after successful cap-and-trade systems, particularly California's.[4] The program establishes a declining cap on statewide emissions. Major emitters must purchase allowances for each ton of carbon dioxide they emit. However, unlike traditional cap-and-trade systems, Washington's Cap-and-Invest Program directs revenue from allowance auctions to specific climate-related investments, including clean energy projects, public transportation improvements, and climate resilience programs.

The program applies to firms emitting 25,000 metric tons or more of CO_2 equivalent an-

[4] California's Cap-and-Trade Program would, in principle, provide an attractive empirical setting because California has a substantially larger economy and a longer-running carbon market. However, California's program began operating in 2013, whereas our firm-level corporate biodiversity footprint (CBF) data begin only in 2018. This prevents us from constructing a sufficiently long pre-treatment period for a difference-in-differences analysis of the California program. Alternative measures, such as satellite-based biodiversity indicators, may provide longer historical coverage to study the California's program, but these types of data do not directly capture firm-level biodiversity footprints, which are the focus of our research question. Washington therefore provides a setting in which the policy shock occurs within the available CBF data period and can be linked directly to firm-level biodiversity outcomes.

nually, covering firms across key sectors: transportation (fuel distributors), energy production (electric utilities), industrial manufacturing (e.g., cement and steel plants), and natural gas distribution. Collectively, these firms account for nearly 70 percent of the state's total greenhouse gas emissions, making this initiative a critical tool for addressing climate change at the state level. Small businesses and entities with emissions below the threshold are exempt from the program, ensuring its focus remains on the largest contributors to emissions.

A. Exogeneity

The treatment assignment in our setting is plausibly exogenous. Washington State enacted the CCA as part of a regional policy trend following California's lead, not in response to firm-specific biodiversity outcomes. The 25,000 metric tons CO_2 emissions threshold is predetermined by existing federal reporting requirements under the Greenhouse Gas Reporting Program, reducing concerns about strategic threshold selection.

B. Treatment Timing

Although the Cap-and-Invest Program officially launched in January 2023, we treat the 2021 enactment date of the CCA as the policy shock for our difference-in-differences analysis. The CCA's passage provided credible advance notice of future compliance obligations. Prior research demonstrates that firms begin adjusting environmental strategies—including capital investment, supply chain renegotiation, and compliance system development—upon policy announcement rather than implementation (Fowlie et al., 2016; Dechezleprêtre and Sato, 2017; Shapiro and Walker, 2018). Consistent with this view, our review of firms' SEC filings (Forms 10-K and 10-Q) between 2021 and 2023 indicates that firms explicitly discussed the CCA, assessed its potential implications, and began evaluating compliance and mitigation strategies well before the program became operational. Moreover, using the enactment date allows us to capture these anticipatory responses, which would be missed if we treated 2023 as the start of treatment.

C. Treated Firms

For identifying treated firms, we rely on the Washington State's Cap-and-Invest Program Quarterly CITSS Registrant Reports. These reports provide the official registry of entities holding active accounts in the Compliance Instrument Tracking System Service (CITSS), the platform used to administer the program. Each registrant is assigned a unique CITSS identification number and is classified according to its regulatory status under the program. Importantly, inclusion in the program is not restricted to firms headquartered in Washington State. Rather, the program covers entities that operate regulated facilities within Washington and whose greenhouse gas emissions exceed the applicable regulatory threshold. The reports distinguish five categories of participants: (*i*) *Covered Entities* or *Covered Sources*, which exceed the emissions threshold for mandatory participation; (*ii*) *Opt-in Entities*, which

voluntarily assume compliance obligations despite falling below the mandatory threshold; (*iii*) *General Market Participant-Organizations*, which participate in the allowance market without compliance obligations; (*iv*) *General Market Participant-Individuals*; and (*v*) *General Market Participant-Offset Only* entities, which are permitted to hold offset credits but face restrictions on allowance transactions.

Because the objective of this study is to evaluate the impact of carbon pricing regulation on firms directly subject to the program, as well as firms that may be indirectly affected through their location and economic exposure to Washington State, we define the treatment group to include (*i*) firms identified in the CITSS registry as Covered Entities/Covered Sources, which are subject to mandatory compliance obligations under the Cap-and-Invest Program, and (*ii*) publicly listed firms headquartered in Washington State that do not appear in the CITSS registry.[5]

III. Data

A. Data Sources

Our empirical analysis draws on two primary data sources: CBF metrics from IDL and financial statement data from Compustat. We merge these sources to construct a comprehensive panel dataset spanning years 2018 through 2024.

A.1. Iceberg Data Lab

IDL provides the CBF data to provide investors with a science-based indicator to help them measure and manage their investments' impact on biodiversity. Recent studies using IDL biodiversity data include Garel et al. (2024) and Coqueret et al. (2025). These studies provide a detailed discussion of the advantages of the IDL framework relative to alternative biodiversity measures, including disclosure-based and textual indicators derived from firms' 10-K statements. The CBF reflects the effects of multiple environmental pressures, such as land use, nitrogen deposition, emissions, or the release of toxic compounds, to quantify the biodiversity loss resulting from a firm's annual activities.

An important limitation of the CBF measure is that, as an aggregated and standardized biodiversity footprint, it does not fully capture differences in the ecological importance of the geographic locations in which biodiversity impacts occur. In particular, one unit of biodiversity loss contributes equally to the aggregate footprint regardless of whether the underlying im-

[5] This broader definition is intended to capture both the direct regulatory effects on covered entities and the potential indirect exposure of locally headquartered firms to policy-induced changes in input costs, energy prices, and regional economic conditions. We do not classify Opt-in Entities as treated because their participation is voluntary rather than mandated by the regulation, making their participation potentially endogenous to firm characteristics. Likewise, General Market Participant-Organizations, General Market Participant-Individuals, and General Market Participant-Offset Only entities are excluded because they participate only in allowance or offset markets and are not subject to mandatory emissions compliance obligations under the Cap-and-Invest Program.

pact occurs in, for example, the United States or a region with substantially different species richness, endemism, or conservation value. Thus, two firms with similar CBF values may generate impacts in locations with very different ecological characteristics. Moreover, because the CBF aggregates both direct and indirect impacts associated with firms' operations and value chains, it should be interpreted as a standardized firm-level measure of estimated biodiversity pressure rather than a direct observation of local species loss. These limitations are important when interpreting the magnitude of our estimated treatment effects. Accordingly, our results should be understood as changes in firms' measured corporate biodiversity footprints, rather than as direct estimates of changes in biodiversity at specific geographic locations.

This measure aggregates biodiversity loss from a company's annual operations (encompassing both direct and indirect impacts),[6] considering six impact categories: GHG emissions, land use, resource consumption, air pollution, water pollution, and invasive alien species.[7] Each channel is defined and operationalized as follows (Iceberg Data Lab, 2023):

- **CBF-GHG:** This channel captures biodiversity loss induced by climate change, to which corporate greenhouse gas (GHG) emissions (e.g., carbon dioxide, methane, and nitrous oxide) contribute. It includes emissions across Scope 1, Scope 2, and Scope 3. Scope 1 refers to direct emissions from owned or controlled sources, Scope 2 captures indirect emissions from purchased energy, and Scope 3 encompasses all other indirect emissions occurring along the value chain, including both upstream activities (e.g., purchased goods and services) and downstream activities (e.g., use and end-of-life treatment of sold products).

- **CBF-Land Use:** This channel quantifies biodiversity loss arising from the physical occupation, conversion, or artificialization of land resulting from a firm's direct operations and value chain activities. Land use pressures include deforestation, agricultural expansion, urban development, mining, and infrastructure construction.

- **CBF-Resource Use:** This channel quantifies biodiversity loss associated with the extraction and consumption of natural resources beyond land occupation, including freshwater, minerals, timber, fisheries, and other biological or geological materials. Overextraction reduces resource availability for native species, degrades habitat structure, and disrupts ecosystem processes. For example, excessive freshwater withdrawal de-

[6] An example of a direct effect is mineral extraction by the same firm, while an indirect effect refers to purchasing renewable energy technologies from other companies, which in turn rely on the extraction of mineral materials for renewable energy production.

[7] To measure this loss, the CBF uses the Mean Species Abundance (MSA) concept, which compares the abundance of native species in ecosystems to their abundance in undisturbed environments. The CBF expresses this loss in km^2.MSA. An area with an MSA of 0 percent has completely lost its native biodiversity (or is exclusively colonized by invasive species), whereas one with an MSA of 100 percent is considered equal in biodiversity to an ecosystem undisturbed by human activities and pressures. The CBF expresses a firm's negative impacts on biodiversity in terms of square kilometers of "artificialized" or "denatured" land (i.e., km^2.MSA). For example, a CBF of -100 km^2 means that 10 percent of the original biodiversity has been lost in an area of 1,000 km^2 or that a proportionally smaller amount of biodiversity, 5 percent, has been lost in an area of 2,000 km^2. In this article, we multiply the CBF scores by -1 so that higher values indicate a more negative impact on biodiversity. For details on how CBF is computed please refer to Garel et al. (2024).

pletes river flows and wetlands, while overharvesting of biological resources collapses food webs and reduces population resilience.

- **CBF-Air Pollution:** This channel measures biodiversity loss driven by the emission of airborne pollutants. Key contributors include sulfur dioxide and nitrogen oxides (precursors to acid rain, which damages forests, soils, and aquatic systems), particulate matter (which reduces photosynthetic capacity and contaminates leaf surfaces), and ground-level ozone (which causes phytotoxicity). Air pollution also alters nutrient deposition patterns, thereby shifting plant community composition and ecosystem function.
- **CBF-Water Pollution:** This channel accounts for biodiversity loss resulting from the discharge of pollutants into freshwater and marine ecosystems. Principal contaminants include industrial chemicals, heavy metals, excess nutrients (nitrogen and phosphorus from agricultural runoff or wastewater), and thermal pollution. These pollutants degrade water quality, induce eutrophication and hypoxic conditions, poison aquatic organisms, and impair critical habitats such as coral reefs, wetlands, and riparian zones.
- **CBF-Invasive Alien Species:** This channel captures biodiversity loss resulting from the introduction and spread of non-native species through corporate activities, such as shipping (ballast water and hull fouling), agricultural trade, or unintentional transport of organisms. Invasive alien species outcompete, predate upon, or transmit pathogens to native species, often leading to population declines, local extirpations, and irreversible alterations to ecosystem structure.

[Insert Figure 1 About Here]

For the period 2018 to 2024, IDL reports both the total CBF and channel-specific CBF for 9,510 publicly listed companies worldwide, including 2,415 U.S. firms. As the CBF is highly skewed, we use log-transformed of all CBF components, and winsorize at the 1st and 99th percentiles to mitigate the concern regarding the outliers in our analysis. Figure 1 decomposes the CBF by underlying pressure source for the U.S. firms. Land use is the largest single contributor at 44.2%, followed by GHG emissions (31%) and air pollution (15.8%); water pollution (7.5%) and resource use (1.6%) play comparatively minor roles. Land use and GHG emissions together account for nearly three-quarter of the classified footprint, indicating that land-intensive and carbon-intensive activities are the dominant sources of CBF.

A.2. Compustat

For the control variables, we follow the literature (e.g., Krüger, 2015; Bolton and Kacperczyk, 2021; Garel et al., 2024) and include a set of firm-level annual financial characteristics capturing size, valuation, profitability, capital structure, investment intensity, and firm's growth. Specifically, we control for *Size*, *Book to Market*, *ROA*, *Leverage*, *Capex to Assets*, and *Asset Growth*. *Size* is defined as natural logarithm of the book value of total assets. *Book to Market* is computed as book equity divided by market equity. *ROA* is measured as net income scaled

by total assets, capturing accounting profitability. *Leverage* is defined as total debt over total assets, where total debt is the sum of long-term debt and debt included in current liabilities. *Capex to Assets* is constructed as capital expenditures divided by total assets, capturing firm-level investment intensity. *Asset Growth* is defined as the annual percentage change in total assets. In addition to these, we include *Asset Turnover*, defined as sales divided by total assets, to capture operational efficiency in generating revenues from the firm's asset base. We further control for *Cash Holdings*, measured as natural logarithm of the ratio of cash and short-term investments to total assets, to account for liquidity and internal financing capacity. This inclusion is motivated by Sonter et al. (2020), who show that firms' cash holdings shape environmental performance.

For all 2,415 U.S. firms identified in the IDL dataset, we collect these variables from Compustat.[8] To mitigate the influence of outliers, we winsorize all variables at the 1st and 99th percentiles. Because the IDL and Compustat datasets do not share a common numeric identifier (e.g., GVKEY or CIK) for all firms, we merge the two datasets using fuzzy string matching on company names. Specifically, we employ a Jaro-Winkler string distance algorithm (Jaro, 1989; Winkler, 1990). Since the CBF is not reported for all firms across the full 2018–2024 period, resulting in an unbalanced panel. Consequently, the final merged sample comprises 12,292 firm-year observations.

B. Matching and Covariate Balance

B.1. Matched Control Group

To construct the control group, we match U.S. firms that are not headquartered in Washington State and do not belong to the treated group (as defined in Subsection II.C) to treated firms using propensity score matching (PSM).[9] We employ PSM to ensure that control firms are comparable to treated firms along observable financial characteristics. To account for potential differences in firms' environmental footprints and operating characteristics across industries, we restrict matching to firms within the same industry, defined by the first two digits of the Global Industry Classification Standard (GICS) code. The matching covariates include the following firm-level variables: *Size*, *Book to Market*, *ROA*, *Leverage*, *Capex to Assets*, *Asset Growth*, *Asset Turnover*, and *Cash Holdings*. Additionally, we include the average pre-policy (2018–2020) GHG component of the corporate biodiversity footprint to account for differences in firms' emission levels prior to the policy intervention.

We implement nearest-neighbor matching. Specifically, we specify a 5:1 matching ratio, reflecting the relatively small number of treated firms (70) compared with the large pool of

[8] For firms subjected to Washington State's Cap-and-Invest Program that are not covered in Compustat, we supplement financial information using Capital IQ.

[9] Firms headquartered in Oregon are excluded from the pool of potential control firms because Oregon adopted significant climate-related regulations beginning with Executive Order 20–04 in 2020, which established statewide greenhouse gas reduction targets and initiated the development of emissions-control policies. Excluding Oregon firms helps avoid contamination of the control group by firms potentially exposed to environmental regulations similar in objective to Washington's Cap-and-Invest Program.

potential control firms (2,174).[10] Under this approach, each treated firm is matched with up to five control firms. Using multiple control matches improves the precision of the estimated treatment effects by reducing sampling variance while maintaining satisfactory covariate balance. To ensure high-quality matches, we follow the recommendation of Austin (2011) and impose a caliper equal to 0.2 of standard deviations of the logit of the propensity score. Consequently, control firms whose propensity scores differ from those of treated firms by more than this threshold are excluded from the matching procedure, thereby reducing the likelihood of poor matches and improving the comparability of the matched sample.[11] This restriction ensures that matches are made only within a region of sufficient common support, thereby improving comparability between treated and control firms and reducing bias from dissimilar observations. The matching is performed without replacement, and common support is enforced to exclude observations with non-overlapping propensity score distributions. The final sample, comprising both treated and control firms, consists of 232 firms, of which 44 are treated. The number of treated firms decreases from 70 to 44 after matching because observations with missing values in one or more matching covariates are excluded from the propensity score estimation and matching procedure. This yields an average of approximately 4.27 control firms per treated firm, below the maximum 5:1 ratio due to the caliper restriction and common support enforcement.

[Insert Figure 2 About Here]

Figure 2 presents the density distributions of the estimated propensity scores for the treated and control groups. The distributions show adequate overlap across the range of propensity scores, particularly in the [0.01–0.10] interval. This overlap confirms that the common support assumption holds reasonably well, allowing for effective matching of treated firms with comparable control firms.

Table 1 presents summary statistics for the biodiversity footprint measures and firm characteristics used in the analysis (please refer to Table A1 for variable definitions). The total CBF exhibits substantial cross-sectional variation, with a mean of 3.808 (corresponding to approximately 45.1 km^2.MSA in biodiversity-equivalent loss) and a standard deviation of 3.193, indicating pronounced heterogeneity in firms' aggregate biodiversity impacts. The distribution ranges from −3.584 to 9.698,[12] with a median (3.872) close to the mean, suggesting a relatively balanced central tendency despite the wide dispersion.

[10] We identify 90 Covered Entities/Covered Sources in the Washington State's Cap-and-Invest Program registry. However, for 64 of these firms, CBF or financial data are unavailable in either the IDL dataset or Compustat (or Capital IQ), leaving 26 covered entities with usable data. Two of these covered entities are also publicly listed firms headquartered in Washington State. In addition, Compustat identifies 131 publicly listed firms headquartered in Washington State, but 85 of these firms do not have corresponding CBF data in the IDL dataset. Therefore, 46 Washington-headquartered firms remain in our sample, of which 2 are also identified as Covered Entities/Covered Sources in the CITSS registry, while the remaining 44 are not listed in the registry. Accounting for this overlap, our final treatment group consists of 70 firms, including 26 CITSS-listed Covered Entities/Covered Sources and 44 Washington-headquartered firms not listed in the CITSS registry.

[11] In Subsection VI.B, we implement several alternative matching specifications to assess the robustness of our findings. Specifically, we estimate the models using 5:1 nearest-neighbor matching with calipers of 0.1 and 0.3, as well as 10:1 nearest-neighbor matching with calipers of 0.1, 0.2, and 0.3.

[12] A negative value indicates that the measured biodiversity loss is less than 1 km^2.MSA.

[Insert Table 1 About Here]

The high mean value of CBF-ex-GHG (3.050) suggests that a substantial share of firms' biodiversity impacts arises from environmental pressure channels other than GHG emissions. Additionally, among the individual biodiversity footprint components, GHG emissions, land use, and air pollution exhibit the highest average biodiversity impacts, indicating that these pressures account for a relatively larger share of firms' biodiversity footprints than the remaining components. Specifically, CBF-GHG has a mean (median) value of 2.705 (2.846) with moderate variability (S.D. = 2.772), indicating that GHG-related biodiversity impacts are widespread but relatively less dispersed across firms. CBF-Land Use displays a mean (median) value of 1.843 (2.127) but the high standard deviation among components (S.D. = 4.066), together with a wide range (−9.185 to 9.040), reflecting substantial heterogeneity in land use-related ecological pressures across firms. CBF-Air Pollution shows a mean (median) value of 1.976 (2.218) and moderate dispersion (S.D. = 3.091), suggesting consistently material but uneven air pollution impacts across the sample.

Turning to the weaker components, water pollution and resource use exhibit lower mean contributions to firms' biodiversity footprints but still display substantial variation across firms. CBF-Water Pollution has a mean of −0.741 and a median of −0.503, with a standard deviation of 3.097 and a wide range from −7.908 to 5.084, indicating that although average impacts are relatively limited, there is considerable dispersion driven by a subset of firms with very high water-related pressures. CBF-Resource Use has a negative average (−2.225) corresponding to 0.11 km^2.MSA biodiversity loss which suggests comparatively lower resource use-related impacts for most firms. However, its dispersion is even more pronounced (S.D. = 4.571), with values ranging from −13.027 to 6.188, reflecting strong dispersion.

The biodiversity footprint associated with invasive alien species is available for only 101 firm-year observations, reflecting substantial data limitations in coverage and reporting. Because of this limited sample size and concerns about representativeness, we exclude this measure from the main analysis. Consequently, we omit this component from the total CBF measure.

The summary statistics for firm-level financial characteristics indicate substantial heterogeneity across the sample. *Size* has a mean of 8.204 and a median of 8.027, with values ranging from 4.640 to 12.655, reflecting considerable cross-firm variation in firm size. *Book to Market* is low on average, with a mean (median) value of 0.008 (0.003) and a relatively tight interquartile range of 0.001 to 0.009, indicating that most firms in the sample are growth-oriented and their market value of equity substantially exceeds the book value, although some variation persists across the upper tail. Profitability, measured by *ROA*, exhibits a mean of approximately 0.000 and a median of 0.031, with values ranging from −0.698 to 0.283, highlighting meaningful differences in firm performance, including a non-negligible share of loss-making firms. *Leverage* has a mean of 0.264 and a standard deviation of 0.196, with values ranging from 0 to 1.092,[13] indicating moderate variation in capital structure

[13] The reported minimum and maximum values are calculated after winsorizing the continuous variables at the 1st and 99th percentiles. Leverage values slightly above one are possible when a firm's book debt exceeds its total

across firms, while the median leverage of 0.247 suggests that more than half of the observations exhibit leverage levels below the mean. Investment intensity, captured by *Capex to Assets*, has a mean of 0.038 and a median of 0.026, with relatively concentrated dispersion between 0.014 and 0.048. The majority of firms invest modestly, as the 75th percentile is only 0.048. However, the maximum value of 0.204 indicates that some firms undertake substantial capital investment. The average *Asset Growth* is 0.192, substantially higher than the median of 0.057, indicating a right-skewed distribution driven by a few high-growth firms. The standard deviation is considerable, at 0.477, reflecting substantial variation in growth rates. Operational efficiency, measured by *Asset Turnover*, has a mean of 0.834 and a median of 0.709, suggesting a right-skewed distribution. The standard deviation of 0.602 indicates significant variation in how efficiently firms utilize their assets to generate sales. The 25th percentile is 0.421, while the 75th percentile is 1.116, with a maximum of 3.009, showing that while many firms generate less than one dollar in sales per dollar of assets, a significant minority exhibit considerably higher asset turnover. *Cash Holdings*, measured by the natural logarithm of the ratio of cash and short-term investments to total assets, has a mean of −2.084 (corresponding to approximately $e^{-2.084} = 0.124$) and a median of −1.971.

B.2. Covariate Balance

An acceptable level of covariate balance between treated and control firms is essential to support the credibility of the identifying assumption that the two groups are also comparable along unobserved dimensions and to mitigate concerns about confounding. We follow the approach in Imbens and Rubin (2015) by employing three metrics to assess the overlap in covariate distributions including:

1) *Normalized Difference*: This statistic is defined as the difference in group means standardized by the pooled standard deviation, where the denominator is the square root of the average variance across the treated and control samples. It is expressed as:

$$\Delta_{c,t} = \frac{\mu_t - \mu_c}{\sqrt{(\sigma_t^2 + \sigma_c^2)/2}}$$

2) *Dispersion Difference*: This metric is computed as the logarithm of the ratio of the standard deviations of treated firms to control firms:

$$\Gamma_{c,t} = \log\left(\frac{\sigma_t}{\sigma_c}\right)$$

3) *Coverage Frequency*: This measure assesses the extent of distributional overlap between the treated and control groups by quantifying the share of observations in one group whose covariate values fall within the extreme tails of the corresponding distribution in the other group. It is formally defined as:

assets, for example among highly levered or financially distressed firms.

$$\pi_t^{\alpha} = F_t\left(F_c^{-1}\left(\frac{\alpha}{2}\right)\right) + \left(1 - F_t\left(F_c^{-1}\left(1 - \frac{\alpha}{2}\right)\right)\right)$$

For a specific value of α, π_t^{α} denotes the probability mass of the covariate's distribution in the sample of treated firms that lies outside the quantiles $1 - \alpha/2$ and $\alpha/2$ of that covariate in the sample of control firms. This measure verifies that the support of the covariate distributions overlaps across groups by checking whether the extreme values in one group remain within the range of those observed in the other group.

To assess overall covariate balance between the two samples, we estimate *P-Score* and the log-odds transformation (*Linearized P-Score*) using a logistic regression in which the treatment indicator is modeled as a function of a set of covariates, including *Size*, *Book to Market*, *ROA*, *Leverage*, *Capex to Assets*, *Asset Growth*, *Asset Turnover*, and *Cash Holdings*. By computing overlap diagnostics—namely the standardized difference, dispersion ratio, and coverage frequency of the estimated propensity scores—we evaluate the degree of covariate balance between the treated and control groups.

[Insert Table 2 About Here]

Table 2 presents covariate balance statistics for both the full and matched samples. Prior to matching, the full sample exhibits substantial imbalance across several firm characteristics. Among the financial variables, the largest absolute normalized differences are observed for *Asset Turnover* (Nor Dif = 0.433), *Leverage* (Nor Dif = −0.323), *Cash Holdings* (Nor Dif = 0.292), *Asset Growth* (Nor Dif = 0.242), *Capex to Assets* (Nor Dif = 0.241), and *Size* (Nor Dif = 0.232), all of which exceed the conventional 0.20 threshold in absolute value. By contrast, *Book to Market* (Nor Dif = −0.173) and *ROA* (Nor Dif = −0.111) exhibit relatively smaller differences before matching. Substantial imbalances are also evident in pre-policy CBF measures. The largest normalized difference is observed for *CBF-Invasive Alien Species* (0.590), followed by *total CBF* (0.318), *CBF-ex-GHG* (0.287), *CBF-Air Pollution* (0.284), *CBF-Water Pollution* (0.282), and *CBF-GHG* (0.278). The Log Ratio of STD further reinforces the presence of imbalance, with several variables displaying sizable differences in dispersion between treated and control firms, including *Size* (0.297), *Capex to Assets* (−0.276), *Asset Turnover* (0.288), *total CBF* (0.333), and *CBF-Land Use* (0.354). These imbalances are also reflected in the propensity-score measures. Both the *P-Score* and *Linearized P-Score* display large pre-matching differences, with normalized differences of 0.603 and 0.870, respectively. These findings suggest that treated and control firms differ considerably in their observable financial characteristics and pre-policy environmental footprints, underscoring the importance of matching to improve their comparability.

The PSM procedure substantially improves covariate balance. After matching, all financial and pre-policy CBF variables exhibit absolute normalized differences below 0.20, with most falling below 0.10. Among the financial characteristics, the largest remaining imbalance is observed for *ROA* (Nor Dif = −0.188), followed by *Asset Growth* (0.141), *Capex to Assets* (−0.108), and *Asset Turnover* (0.103). The improvement in balance is substantial across nearly all financial variables. In absolute terms, the normalized difference declines from 0.232

to 0.020 for *Size*; from 0.173 to 0.035 for *Book to Market*; from 0.323 to 0.001 for *Leverage*; from 0.241 to 0.108 for *Capex to Assets*; from 0.242 to 0.141 for *Asset Growth*; from 0.433 to 0.103 for *Asset Turnover*; and from 0.292 to 0.043 for *Cash Holdings*. Notably, matching also substantially reduces differences in pre-policy environmental footprints. The absolute normalized difference declines from 0.318 to 0.057 for *total CBF*, from 0.287 to 0.058 for *CBF-ex-GHG*, and from 0.278 to 0.010 for *CBF-GHG*. Similar improvements are observed for *CBF-Land Use* (from 0.168 to 0.003), *CBF-Resource Use* (from 0.230 to 0.041), *CBF-Air Pollution* (from 0.284 to 0.003), *CBF-Water Pollution* (from 0.282 to 0.021), and *CBF-Invasive Alien Species* (from 0.590 to 0.031). The propensity-score balance also improves considerably: the normalized difference for the *P-Score* decreases from 0.603 to 0.248, while that for the *Linearized P-Score* declines from 0.870 to 0.218. Although both remain slightly above the 0.20 threshold, these reductions indicate a substantial improvement in the similarity of treated and control firms.

Despite these improvements, some differences remain. The only financial covariate for which the absolute normalized difference increases after matching is *ROA*, rising from 0.111 to 0.188, although its post-matching value remains below the 0.20 threshold. The Log Ratio of STD also improves for most variables, particularly *Size* (from 0.297 to 0.132), *Leverage* (from −0.234 to −0.024), and *Cash Holdings* (from −0.190 to −0.019). However, the dispersion imbalance increases for *Capex to Assets*, with the absolute Log Ratio of STD rising from 0.276 to 0.438. Some differences in dispersion also persist for pre-policy CBF measures, including *total CBF* (0.228), *CBF-ex-GHG* (0.224), and *CBF-Land Use* (0.265). The coverage frequencies ($\pi^{0.05}$), which measure the proportion of observations falling outside the opposite group's 95% quantile range, generally indicate improved overlap following matching, although notable deviations from the 5% benchmark remain for some variables, particularly the pre-policy CBF measures. Overall, the results suggest that combining propensity score matching with exact industry matching substantially reduces observable differences between treated and control firms, particularly in their pre-policy environmental footprints, and provides a more comparable sample for the subsequent DiD analysis.

IV. Identification Strategy

We employ a difference-in-differences (DiD) design to estimate the causal effect of Washington State's Cap-and-Invest Program on CBF. This approach allows us to compare changes in outcomes between treated firms and matched control firms before and after the policy shock. By examining differential changes in outcomes over time, we can isolate the policy effect from other concurrent trends. The PSM approach (as described in Subsection III.B.1) further strengthens identification by ensuring that treated and control firms are comparable on observable characteristics, reducing potential selection bias. Our baseline DiD specification takes the following form:

$$y_{it} = \sum_{k \in K} \beta_k \big(\text{Treat}_i \times \mathbf{1}[t = k]\big) + \delta X_{it} + \alpha_i + \gamma_t + \varepsilon_{it} \tag{1}$$

where y_{it} denotes one of the seven CBF outcomes for firm i in year t: total CBF, CBF-ex-GHG, CBF-GHG, CBF-Land Use, CBF-Resource Use, CBF-Air Pollution, and CBF-Water Pollution; Treat_i takes value 1 for treated firms and zero for firms in the control group; K is the set of years including 2018, 2019, 2020, 2022, 2023, and 2024 (2021 is the omitted reference year); X_{it} is the vector of control variables; α_i and γ_t are firm and year fixed effects; and ε_{it} is the error term clustered at the state level, which accounts for serial and cross-sectional correlation in unobserved shocks among firms exposed to the same state-level regulatory environment, consistent with the fact that Cap-and-Invest is implemented as a single, state-level policy shock (Bertrand et al., 2004; Abadie et al., 2023).

Parallel trends is a central identifying assumption in the DiD framework, requiring that, absent the policy intervention, treated and control firms would have evolved along similar trajectories in corporate biodiversity footprints. The event-study specification allows us to assess whether treated and control firms exhibit any differential dynamics in the pre-policy period (2018–2020), relative to the omitted baseline year 2021, which would be indicative of pre-existing divergence. We also complement these diagnostics with placebo timing tests and alternative matching specifications in the robustness analysis (Section VI) to further examine whether the estimated effects are sensitive to spurious timing or sample construction. Taken together, this combination of design-based and empirical checks is intended to support the credibility of the parallel trends assumption and validity of our DiD analysis, which remains fundamental for causal interpretation of the estimated treatment effects.

A. Heterogeneous Treatment Effects across Firm Characteristics

To further investigate the mechanisms underlying firms' responses to the Cap-and-Invest Program, we examine whether the effects of carbon pricing on corporate biodiversity footprints vary across firm characteristics. Specifically, we develop four hypotheses focusing on firms' pre-policy GHG-related biodiversity exposure, capital investment intensity, asset growth, and firm size.

A.1. Initial GHG-Related Biodiversity Exposure

Firms' responses to carbon pricing may depend on their initial exposure to GHG-related environmental pressures. Firms with higher pre-policy GHG-related biodiversity footprints may face greater regulatory costs and stronger incentives to adjust their production activities in response to carbon pricing. Moreover, these firms may have greater opportunities for environmental improvements, as their initially higher levels of environmental pressure leave more room for reductions through energy efficiency improvements, technological upgrades, and the adoption of cleaner production processes. In contrast, firms with relatively low initial GHG-related biodiversity footprints may already employ more efficient technologies or less carbon-intensive production processes, leaving fewer opportunities for further improvements. Therefore, we expect firms with higher initial GHG-related biodiversity footprints to respond more strongly to carbon pricing and achieve greater reductions in their corporate biodiversity

footprints.

Hypothesis 1 (H1): Firms with higher pre-policy GHG-related biodiversity footprints respond more strongly to carbon pricing, achieving greater reductions in their corporate biodiversity footprints.

A.2. Capital Investment Intensity

Capital investment may influence firms' ability to respond to carbon pricing by facilitating adjustments in production technologies and operational processes. Firms with higher capital investment intensity may be better positioned to replace outdated equipment, adopt cleaner technologies, and improve the efficiency of their production activities. In particular, firms undertaking greater capital expenditures may have more opportunities to incorporate environmental improvements into their investment decisions rather than relying solely on adjustments to existing production facilities. Such investments may help firms reduce the environmental pressures associated with their operations and supply chains. Although capital expenditures may also support production expansion, we expect the technological upgrading and efficiency improvements associated with greater investment intensity to facilitate firms' responses to carbon pricing. Consequently, firms with higher capital investment intensity are expected to achieve larger reductions in their corporate biodiversity footprints.

Hypothesis 2 (H2): Greater capital investment intensity enhances firms' ability to reduce their biodiversity footprints in response to carbon pricing.

A.3. Growth-Driven Expansion

Firms' growth trajectories may also influence their environmental responses to carbon pricing. Rapidly growing firms typically expand their production capacity, increase their demand for energy and intermediate inputs, and undertake additional business activities. These expansions may generate greater environmental pressures, potentially offsetting improvements achieved through cleaner production technologies and more efficient operations. Although growing firms may have opportunities to incorporate environmentally efficient technologies into their expanding asset base, the additional resource requirements and production activities associated with growth may limit the extent to which carbon pricing translates into reductions in corporate biodiversity footprints. In contrast, firms experiencing slower growth may be better able to translate improvements in production efficiency into reductions in their overall environmental pressures. Therefore, we expect growth-related scale effects to weaken the environmental benefits of carbon pricing.

Hypothesis 3 (H3): Higher asset growth weakens the biodiversity footprint reductions associated with carbon pricing, as expansion-related environmental pressures may offset improvements in production efficiency.

A.4. Firm Size and Organizational Flexibility

Firm size may shape firms' responses to carbon pricing through differences in the scale and diversity of their operations. Larger firms often engage in a wider range of production activities and rely on more extensive supply chains, generating biodiversity pressures from multiple sources. Although carbon pricing may encourage these firms to adopt cleaner technologies and improve energy efficiency, such adjustments may not lead to proportional reductions in non-GHG environmental pressures arising from other production processes and supply-chain activities. Consequently, the environmental co-benefits of carbon pricing may be less pronounced among larger firms. In contrast, smaller firms may have more concentrated operations, where improvements in production efficiency and the adoption of cleaner technologies affect a greater proportion of their activities, potentially resulting in stronger reductions in non-GHG biodiversity pressures. We therefore expect larger firms to experience weaker reductions in non-GHG corporate biodiversity footprints following carbon pricing.

Hypothesis 4 (H4): The reductions in corporate biodiversity footprints associated with carbon pricing are less pronounced among larger firms, as their more diversified operations may limit the broader environmental benefits of carbon-focused adjustments.

To test these hypotheses, we extend our baseline difference-in-differences framework by introducing interaction terms between treatment status, year indicators, and the corresponding firm-level characteristics as in Equation (2):

$$y_{it} = \sum_{k \in K} \beta_k \big(\text{Treat}_i \times \mathbf{1}[t = k]\big) + \sum_{k \in K} \theta_k \big(\text{Treat}_i \times Z_{it}^{(j)} \times \mathbf{1}[t = k]\big) + \delta X_{it} + \alpha_i + \gamma_t + \varepsilon_{it}. \quad (2)$$

where $Z_{it}^{(j)}$ denotes the moderator corresponding to hypothesis j. We estimate Equation (2) separately for each of the four moderators: pre-treatment Ln(*CBF-GHG*), *Capex to Assets*, and *Asset Growth*, and *Size*. The coefficient θ_k captures how the treatment effect in event year k varies with the corresponding firm characteristic. These interactions allow us to examine whether the magnitude and direction of treatment effects vary systematically with firms' initial environmental exposure, investment intensity, growth, and organizational scale.

For *Capex to Assets*, *Asset Growth*, and *Size*, the moderator enters the triple interaction using its observed firm-year value. For the emissions-related moderator, however, we do not use the contemporaneous value of Ln(*CBF-GHG*). Since Ln(*CBF-GHG*) is itself one of the dependent variables in our outcome set, using its contemporaneous value as a moderator would mechanically place the same variable on both sides of the regression when the outcome is Ln(*CBF-GHG*). To avoid this problem, we construct a predetermined firm-level GHG moderator using the firm's average Ln(*CBF-GHG*) over the pre-treatment period:

$$Z_i^{GHG} = \frac{1}{T_0} \sum_{t \in \mathcal{T}_0} Ln(\textit{CBF-GHG})_{it}$$

where $\mathcal{T}_0$ denotes the pre-treatment years and T_0 is the number of available pre-treatment

observations for firm i. This variable captures firms' initial GHG-related biodiversity pressure before the policy shock. In specifications using this moderator, $Z_{it}^{(j)}$ in Equation (2) is replaced by the time-invariant baseline measure Z_i^{GHG}.

The parallel trends assumption in the heterogeneous specification is assessed by testing whether the pre-treatment triple-interaction coefficients, θ_{2018}, θ_{2019}, and θ_{2020}, are jointly equal to zero. Failure to reject this null hypothesis indicates that the moderator does not predict differential pre-treatment trends between treated and control firms prior to the introduction of the policy.

V. Results

In this section, we discuss the results of estimation of Equation (1) for exploring the impact of Washington State's Cap-and-Invest Program on seven CBF outcomes: total CBF, CBF-ex-GHG, CBF-GHG, CBF-Land Use, CBF-Resource Use, CBF-Air Pollution, and CBF-Water Pollution. The coefficients of interest are the interaction terms between treatment status and year dummies, with 2021 (the enactment year) as the omitted reference category. Pre-treatment coefficients (2018–2020) test the parallel trends assumption, while post-treatment coefficients (2022–2024) capture the dynamic treatment effects.

As presented in Table 3,[14] the pre-treatment coefficients are small in magnitude and statistically insignificant across all specifications, providing support for the parallel trends assumption. This is further supported by formal Wald/F tests of joint significance of the pre-treatment coefficients (2018–2020). For total CBF ($F = 0.387$, $p = 0.762$), CBF-ex-GHG ($F = 0.611$, $p = 0.608$), CBF-GHG ($F = 0.392$, $p = 0.759$), CBF-Land Use ($F = 0.079$, $p = 0.972$), CBF-Resource Use ($F = 0.613$, $p = 0.607$), CBF-Air Pollution ($F = 0.709$, $p = 0.546$), and CBF-Water Pollution ($F = 0.270$, $p = 0.847$), we fail to reject the null hypothesis that all pre-treatment coefficients are jointly equal to zero. These findings are consistent with the parallel trends assumption across the seven CBF outcomes.

[Insert Table 3 About Here]

Following the 2021 enactment of the Climate Commitment Act, we observe significant reductions in firms' overall biodiversity footprints during the early post-treatment period. For total CBF (Column 1), the treatment effects are negative and statistically significant in 2022 (−0.21) and 2023 (−0.24), both at the 1% level. However, the coefficient for 2024 becomes positive (0.05) and statistically insignificant, suggesting that the initial reductions

[14] The high adjusted R^2 values in the baseline specifications, ranging from 0.96 to 0.97, mainly reflect the inclusion of firm fixed effects. These fixed effects absorb persistent cross-sectional differences across firms, such as industry affiliation, location, business model, production technology, and scale-related biodiversity intensity. This interpretation is supported by an auxiliary specification in which firm fixed effects are replaced with a treated-group dummy while keeping year fixed effects, controls, and event-study interactions. In that specification, the adjusted R^2 falls substantially, ranging from 0.30 to 0.64 across outcomes. This comparison indicates that the high explanatory power of the baseline model is driven primarily by time-invariant firm heterogeneity rather than by the treatment variables themselves. This pattern is expected in firm-level panel data, as biodiversity footprints differ substantially across firms but tend to be relatively persistent within firms over time.

are not sustained through the final year of the sample. Despite this, the aggregated post-treatment effect across 2022–2024, computed as the inverse-variance-weighted average of the corresponding yearly coefficients, remains negative and highly statistically significant ($ATT = -0.21$, significant at the 1% level). This indicates an overall reduction in total CBF over the post-treatment period, driven primarily by the significant effects observed in 2022 and 2023. In economic terms, the estimated aggregate effect corresponds to an approximately 19.0% reduction in firms' biodiversity footprints ($e^{-0.21} - 1$). Using the previously reported pre-treatment mean of 35.9 $km^2 \cdot$MSA as a reference, this implies a decline to approximately 29.1 $km^2 \cdot$MSA. While the estimated effect represents a relatively modest shift in distributional terms (approximately 0.07 standard deviations of log CBF), it suggests a meaningful reduction in environmental impacts during the early years following the enactment of the Climate Commitment Act.

The decomposition of total CBF into its underlying environmental components indicates that the effect of carbon pricing is not similar among different biodiversity components. Starting with the aggregate decomposition, CBF-ex-GHG declines significantly following the policy (Column 2), with coefficients of −0.28 and −0.32 in 2022 and 2023, respectively, both significant at the 1% level. However, the coefficient becomes positive (0.02) and statistically insignificant in 2024, suggesting that these reductions are not sustained through the final year of the sample. These findings indicate that the initial reduction in overall biodiversity footprint is not solely attributable to GHG emissions but also reflects improvements across non-carbon environmental dimensions. Consistent with this pattern, CBF-GHG declines by 0.18 in both 2022 and 2023 (Column 3), with both coefficients significant at the 1% level, indicating meaningful reductions in carbon-related environmental impacts. However, the effect becomes positive (0.12), but statistically insignificant in 2024, suggesting that the initial improvements in GHG-related biodiversity footprints also weaken over time.

Turning to the decomposition across non-GHG components, we observe widespread and statistically significant reductions. The air pollution-related CBF decreases by 0.22 in 2022 and 0.23 in 2023 (both significant at 1% levels, respectively, as depicted in Column 6), with the effect remaining negative and statistically significant in 2024 (−0.68, 1% level). This pattern is consistent with the idea that carbon abatement and air pollution abatement are often complementary. Reductions in fossil-fuel combustion, improvements in energy efficiency, and shifts toward cleaner inputs can lower not only GHG emissions but also local air pollutants generated by the same production activities. The persistence and strengthening of the air pollution effect in 2024 may reflect gradual adjustment, as firms implement operational changes, cleaner technologies, or compliance-related investments over time. Similarly, water pollution-related CBF (Column 7) decreases by 0.22, 0.25, and 0.72 in 2022, 2023, and 2024, respectively, with all reductions statistically significant at the 1% level, indicating sustained and increasingly pronounced improvements in water-related environmental outcomes. This may reflect process-level adjustments that reduce waste streams, chemical discharge, or water-intensive production practices.

For land use and resource use, the estimated effects exhibit different dynamics. The land

use-related CBF (Column 4) declines significantly in 2022 and 2023, with coefficients of −0.37 and −0.36, respectively, both significant at the 1% level. However, the effect reverses in 2024, becoming positive (0.29) and statistically significant at the 5% level. This pattern suggests that the initial reductions in land use-related biodiversity impacts are not sustained and may be followed by an increase in land-related environmental pressures as firms adjust to the implementation of carbon pricing.

One possible explanation for this reversal relates to firms' gradual adjustments in production activities and geographical operations in response to the policy. Following the enactment of the Climate Commitment Act in 2021, firms may have initially responded to anticipated compliance costs by improving operational efficiency or limiting production expansion, contributing to the reductions observed in 2022 and 2023. However, as the Cap-and-Invest Program became operational in 2023 and firms began facing actual carbon compliance costs, their adjustment strategies may have shifted toward more substantial changes in production and investment decisions. In particular, firms with geographically diversified operations, such as CITSS-listed firms headquartered outside Washington State,[15] may have greater flexibility and stronger incentives to relocate or expand production facilities in states with less stringent carbon regulations, potentially shifting land-intensive activities outside Washington. Such geographical reallocation could increase firms' overall land use-related biodiversity footprints if production is shifted to locations with greater land-use intensity or more environmentally sensitive ecosystems. This explanation is consistent with the literature on environmental regulation and firm location decisions, including Becker and Henderson (2000), Copeland and Taylor (2004), Levinson and Taylor (2008), Dechezleprêtre and Sato (2017), and Dechezleprêtre et al. (2022), which highlights how differences in environmental regulatory stringency can influence the geographical distribution of pollution-intensive economic activities.

A further possible channel concerns changes in firms' demand for infrastructure and resource-intensive inputs. As discussed by Vanden Eynde and Vargas (2025), mineral extraction can be associated with land degradation and other environmental impacts. If policy-induced investment or supply-chain changes increase firms' reliance on such activities, their indirect land use-related footprints could also rise. However, our data do not identify the locations of new facilities or suppliers, so neither geographical relocation nor extractive supply-chain adjustment can be established as the cause of the estimated reversal.

Resource Use (Column 5) presents a notable deviation from the overall pattern. While the treatment effect is negative and statistically significant in 2022 (−0.34, 1% level), it reverses in subsequent years. Specifically, the coefficient becomes positive in 2023 (0.29) and increases sharply in 2024 (4.90), both significant at the 1% level, indicating a substantial increase in resource use-related biodiversity pressures during the later post-treatment period. This pro-

[15] These firms have operations in Washington State despite being headquartered elsewhere, suggesting a degree of geographical diversification in their business activities. We further investigate this possibility by restricting the treated sample to CITSS-listed firms headquartered outside Washington State. We find that these firms exhibit higher land use-related CBF than firms headquartered in Washington, consistent with the possibility of geographical reallocation of production activities. The results are not presented in the paper, but available upon request.

nounced reversal contrasts with the generally negative effects observed for other biodiversity footprint components. Consistent with these dynamics, the aggregated post-treatment effect is positive and highly statistically significant (ATT = 0.97, significant at the 1% level), suggesting that, on average, the policy is associated with an increase rather than a reduction in resource use-related biodiversity impacts.

[Insert Figure 3 About Here]

The increase in resource use-related biodiversity pressures should be interpreted in the context of its relatively small contribution to firms' overall biodiversity footprints. As shown in Figure 1, resource use accounts for approximately 1.6% of total CBF among U.S. firms, considerably less than land use and GHG emissions. Therefore, although the estimated increase in resource use is statistically significant and substantial in relative terms, it does not necessarily imply a comparably large deterioration in firms' overall biodiversity footprints. Indeed, our results indicate that total CBF declines significantly in the initial post-treatment years, while the estimated effect in 2024 is not statistically significant. This distinction suggests that the increase in resource use represents a potential trade-off in a particular environmental dimension rather than evidence that carbon pricing increases overall biodiversity loss.

Several competing mechanisms may explain the observed pattern. On the one hand, carbon pricing may encourage firms to improve energy and material efficiency, reduce fossil-fuel consumption, and adopt cleaner production processes, thereby lowering pressures associated with natural resource extraction and land use. On the other hand, adjustments toward low-carbon production may require additional investments in equipment, infrastructure, and alternative inputs, some of which are more material-intensive. For example, electrification, renewable energy systems, and battery technologies depend on minerals such as copper, lithium, nickel, and rare earth elements (IEA, 2021; Sonter et al., 2020). As discussed by Vanden Eynde and Vargas (2025), the green transition can increase demand for these resources and create additional environmental pressures associated with their extraction. In response to carbon pricing, firms may increase their purchases of such technologies or change their suppliers, potentially increasing the resource-use pressures associated with their supply chains. Since the CBF framework captures both direct and indirect environmental impacts, these changes may increase firms' measured resource-related footprints even when resource extraction occurs outside Washington State.

Additional evidence further supports the interpretation that changes in firms' supply chains may contribute to the observed increase in resource-related biodiversity pressures. Figure 3 shows that more than 60% of treated firms experience an increase in their resource use-related footprints following the policy, indicating that this pattern is widespread rather than confined to a small number of firms. Furthermore, Figure 4 reveals a positive and statistically significant treatment effect on upstream Scope 3 GHG-related biodiversity footprints, suggesting an increase in environmental pressures associated with purchased inputs and supply-chain activities. Taken together, these findings provide complementary evidence consistent with firms adjusting their production and sourcing decisions in response to carbon

pricing, potentially shifting environmental pressures toward upstream supply-chain activities. This interpretation highlights how a regulation targeting GHG emissions may have broader consequences for firms' biodiversity footprints beyond their direct operations.

[Insert Figure 4 About Here]

The control variables exhibit several notable relationships with CBF outcomes. Firm size is positively and significantly associated with all CBF components, with coefficients ranging from 0.88 to 1.04 (all significant at the 1% level), indicating that increases in firm size are associated with larger biodiversity footprints, consistent with Garel et al. (2024). Additionally, consistent with Garel et al. (2024), we find that *Asset Growth* is consistently negative and statistically significant across all CBF components, with coefficients ranging from approximately −0.08 to −0.24 (significant at the 5% or 10% level). This pattern indicates that firms experiencing higher growth in total assets tend to exhibit lower biodiversity footprint measures. One plausible interpretation is that rapidly expanding firms may be allocating incremental capital toward more efficient production technologies, cleaner inputs, or improved environmental management systems, thereby decoupling growth from biodiversity-related externalities. Alternatively, this result may reflect changes in firms' asset composition, whereby newly acquired assets incorporate more advanced technologies and stricter environmental standards relative to existing capital stock.

Unlike the previous findings, *ROA* does not exhibit a statistically significant association with any CBF component, suggesting that changes in profitability are not systematically related to biodiversity footprints after controlling for other firm characteristics and fixed effects. *Leverage* is positively associated with all CBF measures, with coefficients ranging from 0.31 to 0.50, although the relationship is statistically significant only for CBF-ex-GHG (10% level) and CBF-Water Pollution (5% level). This provides limited evidence that increases in financial leverage are associated with greater biodiversity-related pressures. In contrast, *Capex to Assets* exhibits a positive association with biodiversity footprints, with coefficients ranging from 1.18 to 3.10 and statistically significant effects in six of the seven specifications. This finding suggests that higher capital investment intensity may be associated with increased environmental pressures, potentially reflecting investments in physical assets, production capacity, and infrastructure that require additional natural resources. Finally, operational efficiency, measured by *Asset Turnover*, is positively and statistically significantly associated with all CBF components, with coefficients ranging from 0.70 to 1.09 (all significant at the 1% level). This indicates that firms generating higher revenues per unit of assets tend to exhibit greater biodiversity footprints, potentially reflecting more intensive utilization of existing production capacity.

A. *Heterogeneous Treatment Effects*

In this subsection, we test the four heterogeneity hypotheses developed in Subsection IV.A. Specifically, we examine whether treatment effects vary with firms' pre-treatment Ln(*CBF-GHG*), *Capex to Assets*, *Asset Growth*, and *Size*.

[Insert Table 4 About Here]

Table 4 presents the results across the seven CBF outcomes. The *Pre* columns report the joint F-tests of the pre-treatment triple-interaction coefficients, while the *Post* columns report the aggregated heterogeneous treatment effects over 2022–2024. Among the 28 specifications, the pre-trend tests fail to reject the null of no differential pre-treatment trends in 11 cases at the 5% significance level. Although these results are consistent with the parallel trends assumption for the corresponding specifications, they do not establish its validity. We therefore focus primarily on specifications without statistically significant differential pre-trends and interpret the remaining estimates with caution.

The findings provide support for *Hypothesis 1*, which predicts that firms with higher initial GHG-related biodiversity footprints respond more strongly to carbon pricing. For the CBF-GHG outcome, the pre-treatment Ln(*CBF-GHG*) moderator exhibits no statistically significant differential pre-trends ($F = 0.87$, $p = 0.458$). The corresponding post-treatment interaction coefficient is negative and statistically significant (-0.029, $p < 0.01$), indicating that firms with greater pre-policy GHG-related biodiversity footprints experience larger reductions in the GHG component following the policy. This finding is consistent with the proposed initial-exposure mechanism: firms with higher baseline GHG-related pressures may face stronger incentives to adjust their production activities and have greater opportunities for emissions abatement. However, the corresponding interaction for total CBF, although negative (-0.015), is not statistically significant ($p = 0.107$), despite the absence of significant differential pre-trends ($F = 1.53$, $p = 0.204$). Thus, the evidence supports stronger GHG-related improvements among initially more exposed firms, but does not establish a similar relationship for their overall biodiversity footprints. For the remaining CBF components, the pre-trend tests reject the null, limiting the interpretation of their post-treatment interaction coefficients.

The results also provide evidence consistent with *Hypothesis 2*, which emphasizes the role of capital investment intensity in firms' environmental adjustments. For total CBF, the pre-trend test is statistically insignificant ($F = 1.85$, $p = 0.136$), while the post-treatment interaction coefficient for *Capex to Assets* is negative and significant (-2.74, $p = 0.011$). A similar pattern emerges for CBF-GHG, where the pre-trend test does not reject the null ($F = 0.81$, $p = 0.487$) and the post-treatment coefficient is negative and significant (-2.19, $p = 0.004$). These findings suggest that firms with higher capital investment intensity experience greater reductions in both their overall and GHG-related biodiversity footprints following carbon pricing. One possible explanation is that capital investment facilitates the replacement of carbon-intensive equipment, technological upgrading, and improvements in production efficiency, allowing firms to respond more effectively to environmental regulation.

Turning to *Hypothesis 3*, the results suggest that firms' growth dynamics may weaken the environmental benefits of carbon pricing. For CBF-GHG, the pre-trend test for *Asset Growth* is statistically insignificant ($F = 1.66$, $p = 0.174$), while the aggregated post-treatment interaction coefficient is positive and statistically significant (0.41, $p = 0.031$). Since the baseline treatment effects on CBF-GHG are negative in the initial post-policy years, the positive in-

teraction indicates that faster-growing firms experience smaller reductions in GHG-related biodiversity pressures than firms with lower asset growth. This finding is consistent with the growth-driven expansion mechanism, whereby increases in production capacity, energy consumption, and demand for intermediate inputs may partially offset environmental improvements associated with carbon pricing. For CBF-Air Pollution, the post-treatment coefficient is also positive and statistically significant (2.31, $p < 0.01$). However, its pre-trend test is marginally significant at the 10% level ($F = 2.30$, $p = 0.075$), suggesting that this result should be interpreted cautiously. For the other CBF components, significant differential pre-trends prevent us from drawing reliable conclusions about heterogeneous treatment effects associated with asset growth.

Finally, the evidence for Hypothesis 4 suggests that firm size may influence the environmental response to carbon pricing, particularly for non-GHG biodiversity pressures. For CBF-ex-GHG, the pre-trend test does not reject the null ($F = 1.76$, $p = 0.154$), while the post-treatment interaction coefficient for Size is positive and statistically significant (0.041, $p = 0.007$). Similar results are observed for CBF-Air Pollution and CBF-Water Pollution, where the pre-trend tests are statistically insignificant ($F = 1.03$, $p = 0.381$; and $F = 0.83$, $p = 0.480$, respectively), while the post-treatment interaction coefficients are positive and significant (0.029, $p = 0.036$; and 0.029, $p = 0.048$, respectively). These findings indicate that larger firms experience weaker reductions in these non-GHG biodiversity pressures following the policy. One possible explanation is that larger firms operate across a wider range of production activities and supply chains, generating environmental pressures from multiple sources that are not necessarily directly linked to GHG emissions. Consequently, adjustments undertaken in response to carbon pricing may reduce emissions from carbon-intensive activities without generating proportional improvements in other environmental dimensions. In contrast, smaller firms with more concentrated operations may experience stronger environmental co-benefits when adopting cleaner technologies or improving production efficiency. This mechanism is consistent with the weaker reductions in non-GHG biodiversity pressures observed among larger firms. Moreover, the interactions for total CBF and CBF-GHG are not statistically significant ($p = 0.512$ and $p = 0.502$, respectively), despite the absence of significant differential pre-trends. The results therefore provide support for size-related heterogeneity in selected non-GHG components rather than in firms' overall biodiversity footprints. For land use and resource use, the pre-trend tests reject the null, limiting the interpretation of their post-treatment coefficients.

Overall, the heterogeneity analysis provides evidence that firms' responses to carbon pricing depend on their initial environmental exposure, investment intensity, growth dynamics, and organizational scale. Firms with higher pre-policy GHG-related biodiversity footprints and greater capital investment intensity exhibit stronger reductions in certain biodiversity measures, consistent with greater abatement opportunities and technological adjustment capacity. In contrast, faster-growing firms experience weaker GHG-related improvements, suggesting that expansion-related scale effects may offset some of the environmental benefits of carbon pricing. Larger firms also exhibit weaker reductions in several non-GHG biodiversity

components, potentially because their more diversified operations generate environmental pressures from multiple sources that are not directly linked to GHG emissions. Consequently, carbon-focused adjustments may produce smaller improvements in other environmental dimensions for these firms. However, these relationships are not uniform across all biodiversity pressures. Consequently, the findings provide evidence of heterogeneous responses across particular environmental dimensions rather than a uniform effect of firm characteristics on overall corporate biodiversity footprints.

VI. Robustness Checks

A. Placebo Timing Test

We conduct a placebo timing test to assess the validity of our difference-in-differences identification strategy. Specifically, we re-estimate the baseline specification, as in Equation (1), by assigning a fictitious treatment year prior to the actual enactment of Washington State's Climate Commitment Act and examine the dynamic treatment coefficients across all outcome variables. The objective of this exercise is to test whether the empirical model spuriously generates treatment effects in periods when no policy intervention has occurred.

In this placebo exercise, we set the false treatment timing to 2020, which precedes the actual policy enactment in 2021, and re-estimate the full event-study specification using the same dynamic structure as in the baseline model. This implies that 2018–2019 serve as pre-placebo periods, while 2021–2024 are treated as post-placebo periods, even though no real policy change occurred at the placebo date. A valid DiD design should exhibit no systematic or statistically significant effects in pre-treatment periods, as this would indicate that treated and control firms follow parallel trends prior to the policy shock. Moreover, under a correctly specified model, we should not observe economically meaningful or systematically significant treatment effects immediately following the placebo year (i.e., in 2021), since no actual regulatory intervention took place at that time. Instead, coefficients should remain statistically indistinguishable from zero in both pre-placebo periods and the immediate post-placebo year. Evidence of systematic pre-trends in the pre-placebo period or statistically significant effects immediately following the fictitious treatment year would cast doubt on the validity of our DiD design.

[Insert Figure 5 About Here]

As shown in Figure 5, the placebo estimates provide broadly supportive evidence for the validity of the identification strategy across most CBF outcomes. For the all of specifications, pre-placebo coefficients are statistically indistinguishable from zero and insignificant, consistent with the absence of differential pre-trends. In addition, the interaction term for 2021—corresponding to the first post-placebo period—remains statistically insignificant across all outcomes, indicating that no spurious treatment effects are detected immediately after the

fictitious policy implementation date. This further supports the validity of the placebo design and reinforces the credibility of the parallel trends assumption underlying our baseline specification.

Overall, the placebo exercise provides supportive evidence for the validity of the design, with all outcomes showing no meaningful pre-trend behavior and treatment effects in the main specification emerging only in the actual post-policy period.

B. Alternative Matching Specifications

To assess the sensitivity of our baseline results to the matching procedure, we re-estimate the analysis employing several alternative propensity score matching configurations. These include 5:1 nearest-neighbor matching with calipers set to 0.1 and 0.3, alongside 10:1 nearest-neighbor matching with calipers of 0.1, 0.2, and 0.3. This battery of estimates serves as a robustness check against dependence on the specific matching design. Should our findings be robust—rather than an artifact of the chosen parameters or arbitrary pairings—we would expect the treatment effects to retain their qualitative consistency across all specifications in terms of direction, temporal dynamics, and statistical significance.

[Insert Figure 6 About Here]

Figure 6 presents the dynamic treatment effects under the alternative matching specifications for all seven CBF outcomes. Overall, the results exhibit substantial consistency with the baseline findings, particularly in terms of the direction and timing of the post-treatment effects. The pre-treatment coefficients for 2018–2020 are generally statistically insignificant across most outcomes and matching configurations. However, some exceptions emerge, particularly in 2018, where the coefficients for total CBF and CBF-Air Pollution become significantly negative under certain 10:1 matching specifications. Although the pre-treatment coefficients for 2019 and 2020 remain largely insignificant, these exceptions suggest that the parallel trends assumption may be more sensitive to the matching procedure for certain outcomes.

Turning to the post-treatment period, the results for total CBF and CBF-ex-GHG remain broadly consistent across the alternative matching specifications. Both outcomes exhibit negative and statistically significant treatment effects in 2022 and 2023, with relatively similar magnitudes across the different matching ratios and calipers. However, the coefficients become positive and statistically insignificant in 2024 under all alternative specifications. This pattern is consistent with the baseline findings, suggesting that the initial reductions in firms' overall biodiversity footprints are not sustained through the final year of the sample. Similarly, the GHG-related CBF exhibits significant reductions in 2022 and 2023 across the alternative specifications. In 2024, however, the coefficients become positive, and some are statistically significant under tighter caliper restrictions. This suggests that the weakening of the initial GHG-related improvements is robust to alternative matching procedures, although the statistical significance of the subsequent reversal depends on the matching specification.

For the non-GHG components, CBF-Air Pollution and CBF-Water Pollution exhibit particularly consistent post-treatment patterns. Both components show negative and statistically

significant coefficients in 2022 and 2023, with the reductions becoming substantially larger in 2024. The estimated effects remain negative and statistically significant across all alternative matching specifications in the post-treatment period. This consistency suggests that the reductions in air and water pollution-related biodiversity pressures are not driven by the particular matching ratio or caliper selected in the baseline analysis.

The results for CBF-Land Use and CBF-Resource Use also largely preserve the dynamic reversals observed in the baseline estimates. For CBF-Land Use, the treatment effects remain negative and statistically significant in 2022 and 2023 across all alternative specifications. In contrast, the coefficients become positive in 2024 and are statistically significant in most specifications. This finding indicates that the reversal in land use-related biodiversity pressures is not specific to the baseline matching procedure.

Similarly, CBF-Resource Use exhibits negative and statistically significant treatment effects in 2022, followed by positive coefficients in 2023 and a substantial increase in 2024. Although the statistical significance of the 2023 estimates varies across matching specifications, the 2024 coefficients remain positive and highly statistically significant under all configurations. Nevertheless, the magnitude of the 2024 effect varies considerably across matching specifications, with estimates ranging approximately from 5 to 8 log points. This variation indicates that, while the direction and timing of the resource-use reversal are robust, its estimated magnitude is relatively sensitive to the composition of the matched sample. Accordingly, the size of this effect should be interpreted with caution.

Overall, the alternative matching specifications provide broadly supportive evidence for the main findings. In particular, the initial reductions in total CBF and CBF-GHG, the persistent improvements in air and water pollution-related biodiversity footprints, and the subsequent increases in land use and resource use-related pressures are generally preserved across different matching ratios and calipers. These findings suggest that the contrasting environmental effects of carbon pricing are not solely attributable to the baseline matching specification. However, the significant pre-treatment coefficients observed in some specifications, the sensitivity of the 2024 GHG-related estimates, and the variation in the magnitude of the resource-use effect highlight important limitations. Taken together, the results support the qualitative robustness of the main findings while suggesting greater caution regarding the precise magnitude of certain treatment effects.

VII. Conclusion

This paper examines the impact of a market-based carbon pricing regulation on corporate biodiversity outcomes. Using firm-level data on the corporate biodiversity footprint (CBF) from Iceberg Data Lab merged with financial statement information from Compustat, we study Washington State's Cap-and-Invest Program as a quasi-experimental policy shock. Combining a difference-in-differences framework with propensity score matching, we examine how firms respond to carbon pricing across multiple biodiversity pressure categories.

Our findings suggest that carbon pricing has meaningful but heterogeneous effects on

corporate biodiversity footprints. Following the enactment of Washington State's Climate Commitment Act in 2021, total CBF declines significantly in 2022 and 2023, although this reduction is no longer statistically significant in 2024. The decomposition of CBF reveals substantial differences across environmental dimensions. While air and water pollution-related biodiversity pressures decline persistently throughout the post-treatment period, land use and resource use exhibit different dynamics. Both components initially decline following policy enactment, but their effects subsequently reverse. In particular, resource use-related biodiversity pressures increase significantly in 2023 and 2024, while land use-related pressures become significantly positive in 2024. These findings suggest that the environmental benefits of carbon pricing are neither uniform across biodiversity dimensions nor necessarily sustained over time. Although carbon pricing can generate important environmental co-benefits beyond greenhouse gas emissions, it may also be associated with increasing pressures on land and natural resources. These components nevertheless differ considerably in their contribution to aggregate CBF: resource use represents approximately 1.6% of aggregate CBF in the broader U.S. sample, whereas land use accounts for approximately 44.2%. Thus, the large relative increase in resource-use CBF should not be equated with a similarly large increase in total CBF. Overall, these contrasting patterns point to potential trade-offs across biodiversity dimensions rather than a uniform deterioration in firms' environmental performance.

The heterogeneity analysis further indicates that firms' responses to carbon pricing depend on their initial environmental exposure, investment intensity, growth dynamics, and organizational scale. Firms with higher pre-policy GHG-related biodiversity footprints experience larger reductions in the GHG component, consistent with stronger regulatory incentives and greater opportunities for environmental improvement. Similarly, firms with higher capital investment intensity exhibit larger reductions in total and GHG-related biodiversity footprints, suggesting that investment may facilitate technological upgrading and adjustments toward cleaner production. In contrast, faster-growing firms experience weaker reductions in GHG-related biodiversity pressures, potentially reflecting growth-related scale effects that offset improvements in production efficiency. We also find that larger firms exhibit weaker reductions in several non-GHG biodiversity components, particularly air and water pollution, one possible explanation is that because their more diversified operations generate environmental pressures from multiple sources, limiting the extent to which carbon-focused adjustments translate into broader environmental improvements. Nevertheless, these relationships are not uniform across all CBF components, and some specifications exhibit significant differential pre-treatment trends, requiring a cautious interpretation. Overall, the findings suggest that the biodiversity consequences of carbon pricing depend partly on firms' characteristics and their ability to adjust production activities in response to environmental regulation.

Our results carry important implications for environmental policy design. The evidence suggests that carbon pricing can reduce several dimensions of firms' environmental footprints, but carbon pricing alone may not fully address the broader environmental externalities associated with the transition toward a low-carbon economy. In particular, the increasing bio-

diversity pressures associated with resource use and land use highlight the importance of considering potential unintended consequences of climate-focused regulations. These effects may arise from greater demand for material-intensive technologies and infrastructure or from adjustments in firms' production locations and supply chains. Although our analysis does not directly establish these mechanisms, the findings suggest that policies targeting greenhouse gas emissions may need to be complemented by measures addressing other environmental pressures. Policies that promote energy and material efficiency, encourage cleaner production technologies, support recycling and circular production processes, and limit environmentally harmful land conversion may help mitigate such trade-offs. Greater attention to environmental pressures along firms' supply chains and across geographical locations may also improve the alignment between climate mitigation and biodiversity conservation objectives. In this regard, Vanden Eynde and Vargas (2025) highlight the wider economic and governance challenges associated with natural resource extraction during the green transition. Their review suggests a broader rationale for combining climate policy with responsible resource governance, including transparency about mineral sourcing and stronger oversight of extractive activities. These represent potential complementary policy measures, although our data do not allow us to evaluate their effectiveness.

Future research could extend this analysis in several directions. First, the current study is constrained by the limited time coverage of the CBF data, particularly the relatively short period following the implementation of Washington State's Cap-and-Invest Program in 2023. Extending the sample period would allow researchers to examine whether the observed changes in biodiversity pressures persist, intensify, or reverse over longer horizons. A longer historical dataset would also enable the study of other large-scale carbon pricing systems, such as California's cap-and-trade program, which began operating in 2013 and covers a broader set of firms and industries. Second, future research could investigate the mechanisms underlying the observed increases in land use and resource use-related biodiversity pressures. More detailed information on firms' production locations, investment decisions, and supply-chain activities would help determine whether these effects arise from technological substitution, increased demand for natural resources, or the geographical reallocation of production. Linking firm-level footprints with supplier locations, mineral sourcing, and the environmental characteristics of extraction sites would also help identify where indirect biodiversity pressures arise. In a broader context, Vanden Eynde and Vargas (2025) point to the need for more evidence on the environmental and socioeconomic consequences of resource exploitation during the green transition; tracing these links from regulated firms to their upstream activities would be a useful extension of our analysis. Third, future studies could examine alternative forms of carbon pricing and environmental regulation, including carbon taxes, performance-based emissions standards, or hybrid regulatory systems combining price and quantity instruments. Such extensions would help clarify whether the observed biodiversity effects are specific to cap-and-invest designs or generalizable across different regulatory frameworks. Finally, more granular regulatory variation, such as differentiated emissions thresholds and sector-specific compliance requirements, could provide further insights into

how particular policy features shape corporate biodiversity outcomes. Together, these extensions would contribute to a more comprehensive understanding of how climate policy interacts with biodiversity across firms, industries, and institutional settings.

References

Abadie, Alberto, Susan Athey, Guido W Imbens, and Jeffrey M Wooldridge, "When should you adjust standard errors for clustering?," *The Quarterly Journal of Economics*, 2023, *138* (1), 1–35.

Acemoglu, Daron, Philippe Aghion, Leonardo Bursztyn, and David Hemous, "The environment and directed technical change," *American Economic Review*, 2012, *102* (1), 131–166.

Agreement, Paris, "Paris agreement," in "report of the conference of the parties to the United Nations framework convention on climate change (21st session, 2015: Paris). Retrived December," Vol. 4 HeinOnline 2015, p. 2.

Auffhammer, Maximilian and Ryan Kellogg, "Clearing the air? The effects of gasoline content regulation on air quality," *American Economic Review*, 2011, *101* (6), 2687–2722.

Austin, Peter C, "Optimal caliper widths for propensity-score matching when estimating differences in means and differences in proportions in observational studies," *Pharmaceutical Statistics*, 2011, *10* (2), 150–161.

Becker, Randy and Vernon Henderson, "Effects of air quality regulations on polluting industries," *Journal of Political Economy*, 2000, *108* (2), 379–421.

Berman, Eli and Linda TM Bui, "Environmental regulation and productivity: evidence from oil refineries," *Review of Economics and Statistics*, 2001, *83* (3), 498–510.

Bertrand, Marianne, Esther Duflo, and Sendhil Mullainathan, "How much should we trust differences-in-differences estimates?," *The Quarterly Journal of Economics*, 2004, *119* (1), 249–275.

Bohnet, Marc-Philipp, Stefano Carattini, and Suphi Sen, "Carbon pricing and investors' reaction: evidence from Washington State," 2026.

Bolton, Patrick and Marcin Kacperczyk, "Do investors care about carbon risk?," *Journal of Financial Economics*, 2021, *142* (2), 517–549.

Bu, Caiqi and Daqian Shi, "The emission reduction effect of daily penalty policy on firms," *Journal of Environmental Management*, 2021, *294*, 112922.

Bushnell, James B and Catherine D Wolfram, "Enforcement of vintage differentiated regulations: The case of new source review," *Journal of Environmental Economics and Management*, 2012, *64* (2), 137–152.

Calel, Raphael and Antoine Dechezleprêtre, "Environmental policy and directed technological change: evidence from the European carbon market," *Review of Economics and Statistics*, 2016, *98* (1), 173–191.

Cole, Matthew A, "Trade, the pollution haven hypothesis and the environmental Kuznets curve: examining the linkages," *Ecological Economics*, 2004, *48* (1), 71–81.

Copeland, Brian R and M Scott Taylor, "Trade, growth, and the environment," *Journal of Economic Literature*, 2004, *42* (1), 7–71.

Coqueret, Guillaume, Thomas Giroux, and Olivier David Zerbib, "The biodiversity premium," *Ecological Economics*, 2025, *228*, 108435.

Curtis, E Mark, "Who loses under cap-and-trade programs? The labor market effects of the NOx budget trading program," *Review of Economics and Statistics*, 2018, *100* (1), 151–166.

Dechezleprêtre, Antoine and Misato Sato, "The impacts of environmental regulations on competitiveness," *Review of Environmental Economics and Policy*, 2017.

_ , Caterina Gennaioli, Ralf Martin, Mirabelle Muûls, and Thomas Stoerk, "Searching for carbon leaks in multinational companies," *Journal of Environmental Economics and Management*, 2022, *112*, 102601.

Eynde, Oliver Vanden and Juan Vargas, "Climate change, natural resources, and conflict," *Economic Policy*, 2025, *40* (123), 651–677.

Fischer, Carolyn and Richard G Newell, "Environmental and technology policies for climate mitigation," *Journal of Environmental Economics and Management*, 2008, *55* (2), 142–162.

Flammer, Caroline, "Corporate green bonds," *Journal of Financial Economics*, 2021, *142* (2), 499–516.

Fowlie, Meredith, "Emissions trading, electricity restructuring, and investment in pollution abatement," *American Economic Review*, 2010, *100* (3), 837–869.

_ , Mar Reguant, and Stephen P Ryan, "Market-based emissions regulation and industry dynamics," *Journal of Political Economy*, 2016, *124* (1), 249–302.

Garel, Alexandre, Arthur Romec, Zacharias Sautner, and Alexander F Wagner, "Do investors care about biodiversity?," *Review of Finance*, 2024, *28* (4), 1151–1186.

Greenstone, Michael, "Estimating regulation-induced substitution: The effect of the Clean Air Act on water and ground pollution," *American Economic Review*, 2003, *93* (2), 442–448.

Iceberg Data Lab, "Corporate Biodiversity Footprint: Methodological Guide," 2023.

IEA, "The Role of Critical Minerals in Clean Energy Transitions," 2021.

Imbens, Guido W and Donald B Rubin, *Causal inference in statistics, social, and biomedical sciences*, Cambridge University Press, 2015.

IPBES, "Workshop Report on Biodiversity and Pandemics: Workshop Report," 2020.

IPCC, *Climate Change 2014: Mitigation of Climate Change. Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*, Cambridge, United Kingdom and New York, NY, USA: Cambridge University Press, 2014.

Jaro, Matthew A, "Advances in record-linkage methodology as applied to matching the 1985 census of Tampa, Florida," *Journal of the American Statistical Association*, 1989, *84* (406), 414–420.

Krüger, Philipp, "Corporate goodness and shareholder wealth," *Journal of Financial Economics*, 2015, *115* (2), 304–329.

Kunming Declaration, "Kunming Declaration: Declaration from the High-Level Segment of the UN Biodiversity Conference 2020 (Part 1) under the Theme: Ecological Civilization: Building a Shared Future for All Life on Earth," 2021. Adopted at COP15 (Part 1), Kunming, China.

Levinson, Arik and M Scott Taylor, "Unmasking the pollution haven effect," *International Economic Review*, 2008, *49* (1), 223–254.

Li, Panni, Zhongguo Lin, Huibin Du, Tong Feng, and Jian Zuo, "Do environmental taxes reduce air pollution? Evidence from fossil-fuel power plants in China," *Journal of Environmental Management*, 2021, *295*, 113112.

Liang, Yuanning, Ivan J. Rudik, Eric Yongchen Zou, Alison Johnston, Amanda D. Rodewald, and Catherine L. Kling, "Conservation Co-Benefits from Air Pollution Regulation: Evidence from Birds," *Proceedings of the National Academy of Sciences*, 2020, *117* (49), 30900–30906.

Martin, Ralf, Mirabelle Muûls, Laure B De Preux, and Ulrich J Wagner, "Industry compensation under relocation risk: A firm-level analysis of the EU emissions trading scheme," *American Economic Review*, 2014, *104* (8), 2482–2508.

Matsumura, Ella Mae, Rachna Prakash, and Sandra C Vera-Muñoz, "Firm-value effects of carbon emissions and carbon disclosures," *The Accounting Review*, 2014, *89* (2), 695–724.

Montreal Agreement, "COP 15: Nations Adopt Four Goals, 23 Targets for 2030 In Landmark UN Biodiversity Agreement," 2022. Adopted at COP15, Montreal, Canada.

OECD, *Towards Green Growth*, OECD Publishing, 2011.

Porter, Michael E and Claas van der Linde, "Toward a new conception of the environment-competitiveness relationship," *Journal of Economic Perspectives*, 1995, *9* (4), 97–118.

Press, Elizabeth, Thijs Van de Graaf, Martina Lyons, Isaac Elizondo Garcia, Ellipse Rath, Benjamin Gibson et al., "Geopolitics of the energy transition: critical materials," 2023.

Shapiro, Joseph S and Reed Walker, "Why is pollution from US manufacturing declining? The roles of environmental regulation, productivity, and trade," *American Economic Review*, 2018, *108* (12), 3814–3854.

Sonter, Laura J, Marie C Dade, James EM Watson, and Rick K Valenta, "Renewable energy production will exacerbate mining threats to biodiversity," *Nature Communications*, 2020, *11* (1), 4174.

Winkler, William E, "String comparator metrics and enhanced decision rules in the fellegi-sunter model of record linkage.," 1990.

World Bank, "Mobilizing Private Finance for Nature," 2020.

WWF, "Living Planet Report 2024 – A System in Peril," 2024. WWF, Gland, Switzerland.

Zhou, Zoey Yiyuan and Douglas Almond, "Biodiversity Co-Benefits in Carbon Markets? Evidence from Voluntary Offset Projects," *Review of Finance*, 2026, *30* (2), 537–570.

TABLES

Table 1: Summary Statistics

	Mean	S.D.	Min	25%	50%	75%	Max	Obs.
Ln(CBF)	3.808	3.193	-3.584	1.276	3.872	6.345	9.698	1,342
Ln(CBF−ex−GHG)	3.050	3.512	-4.826	0.356	3.091	5.796	9.328	1,342
Ln(CBF−GHG)	2.705	2.772	-4.051	0.583	2.846	4.810	8.469	1,342
Ln(CBF−Land Use)	1.843	4.066	-9.185	-1.064	2.127	4.913	9.040	1,339
Ln(CBF−Resource Use)	-2.225	4.571	-13.027	-6.028	-2.176	1.285	6.188	1,200
Ln(CBF−Air Pollution)	1.976	3.091	-5.629	-0.371	2.218	4.251	7.846	1,342
Ln(CBF−Water Pollution)	-0.741	3.097	-7.908	-3.066	-0.503	1.579	5.084	1,342
Ln(CBF−Invasive Alien Species)	0.238	3.890	-7.935	-2.996	0.468	2.846	7.673	101
Size	8.204	1.708	4.640	6.918	8.027	9.477	12.655	1,340
Book to Market	0.008	0.013	-0.008	0.001	0.003	0.009	0.093	1,248
ROA	-0.000	0.152	-0.698	-0.032	0.031	0.079	0.283	1,340
Leverage	0.264	0.196	0.000	0.097	0.247	0.387	1.092	1,331
Capex to Assets	0.038	0.039	0.000	0.014	0.026	0.048	0.204	1,340
Asset Growth	0.192	0.477	-0.352	-0.014	0.057	0.193	2.787	1,329
Asset Turnover	0.834	0.602	0.012	0.421	0.709	1.116	3.009	1,340
Cash Holdings	-2.084	1.247	-6.911	-2.839	-1.971	-1.087	-0.102	1,340

This table presents summary statistics at the firm-year level of the variables used in the returns analysis. The sample period spans from 2018 to 2024. Please refer to Table A1 for variable definitions.

Table 2: Covariate Balance Test

	Full Sample				Matched Sample (PSM)			
			$\pi^{0.05}$				$\pi^{0.05}$	
	Nor Dif	Log Ratio of STD	Control	Treated	Nor Dif	Log Ratio of STD	Control	Treated
Size	0.2319	0.2969	0.0209	0.1400	0.0198	0.1315	0.0638	0.0682
Book to Market	−0.1728	−0.1207	0.0877	0.0200	−0.0352	0.0674	0.0585	0.0682
ROA	−0.1111	−0.0181	0.0952	0.0400	−0.1879	0.0047	0.0957	0.0455
Leverage	−0.3231	−0.2336	0.1202	0.0400	0.0007	−0.0244	0.0638	0.0682
Capex to Assets	0.2409	−0.2761	0.2311	0.0000	−0.1076	−0.4380	0.1968	0.0000
Asset Growth	0.2420	0.2526	0.1069	0.1000	0.1410	0.1756	0.0851	0.0909
Asset Turnover	0.4327	0.2877	0.1452	0.1200	0.1033	−0.0167	0.1277	0.0227
Cash Holdings	0.2921	−0.1899	0.1957	0.0000	0.0428	−0.0188	0.1862	0.0227
Ln(CBF)	0.3184	0.3333	0.0151	0.2800	0.0571	0.2282	0.0053	0.2045
Ln(CBF-ex-GHG)	0.2874	0.3273	0.0175	0.2800	0.0584	0.2241	0.0213	0.2500
Ln(CBF-GHG)	0.2782	0.2913	0.0180	0.2000	−0.0098	0.1824	0.0053	0.1591
Ln(CBF-Land Use)	0.1676	0.3542	0.0245	0.1800	−0.0025	0.2648	0.0000	0.1591
Ln(CBF-Resource Use)	0.2295	0.1990	0.0364	0.1739	−0.0414	0.1132	0.0465	0.1000
Ln(CBF-Air Pollution)	0.2843	0.2623	0.0273	0.2400	0.0032	0.1604	0.0106	0.1818
Ln(CBF-Water Pollution)	0.2815	0.2820	0.0181	0.2600	0.0214	0.1823	0.0053	0.2045
Ln(CBF-Invasive Alien Species)	0.5895	0.1509	0.4222	0.1667	−0.0314	0.1202	0.2308	0.2000
P-Score	0.6026	1.0988	0.2625	0.1400	0.2480	0.2796	0.0372	0.0909
Linearized P-Score	0.8701	0.1440	0.2625	0.1400	0.2184	0.1220	0.0372	0.0909

This table presents the covariate balance between treated and control firms before and after propensity score matching. The matching procedure uses firm financial characteristics and pre-policy GHG emissions, with exact matching on two-digit GICS sectors and a matching ratio of 1:5 with a caliper of 0.2. All CBF measures are calculated as firm-level averages over the pre-policy period (2018–2020). The overlap measures include normalized differences (Nor Dif), dispersion differences (Log Ratio of STD), and coverage frequency ($\pi^{0.05}$). Propensity scores (P-Score) and linearized propensity scores (Linearized P-Score) are also reported. Please refer to Table A1 for variable definitions.

Table 3: Dynamic Treatment Effects Results

	(1) Total	(2) Ex-GHG	(3) GHG	(4) Land Use	(5) Res. Use	(6) Air Pol.	(7) Water Pol.
Interaction Terms							
Treat × Year (2018)	−0.09	−0.14	0.01	−0.01	−0.09	−0.15	−0.06
	(0.09)	(0.12)	(0.08)	(0.13)	(0.12)	(0.11)	(0.16)
Treat × Year (2019)	−0.10	−0.13	0.01	0.00	−0.11	−0.13	−0.07
	(0.11)	(0.12)	(0.07)	(0.11)	(0.10)	(0.13)	(0.13)
Treat × Year (2020)	−0.01	−0.06	0.03	−0.05	−0.01	−0.03	−0.04
	(0.06)	(0.08)	(0.04)	(0.14)	(0.13)	(0.06)	(0.05)
Treat × Year (2022)	−0.21***	−0.28***	−0.18***	−0.37***	−0.34***	−0.22***	−0.22***
	(0.03)	(0.04)	(0.04)	(0.06)	(0.08)	(0.04)	(0.04)
Treat × Year (2023)	−0.24***	−0.32***	−0.18***	−0.36***	0.29***	−0.23***	−0.25***
	(0.05)	(0.05)	(0.03)	(0.08)	(0.11)	(0.05)	(0.05)
Treat × Year (2024)	0.05	0.02	0.12	0.29**	4.90***	−0.68***	−0.72***
	(0.14)	(0.15)	(0.15)	(0.12)	(0.12)	(0.16)	(0.18)
Control Variables							
Size	0.91***	0.95***	0.88***	1.04***	0.96***	0.94***	0.91***
	(0.11)	(0.10)	(0.11)	(0.11)	(0.16)	(0.09)	(0.09)
Book to Market	−7.64	−8.58	−6.91	−8.80	−4.18	−11.74	−12.29
	(5.13)	(5.48)	(5.81)	(5.56)	(9.82)	(9.49)	(9.82)
ROA	0.20	0.29	0.07	0.38	−0.01	0.12	0.25
	(0.36)	(0.37)	(0.35)	(0.38)	(0.46)	(0.34)	(0.33)
Leverage	0.36	0.50*	0.31	0.38	0.45	0.34	0.39**
	(0.26)	(0.27)	(0.23)	(0.45)	(0.40)	(0.24)	(0.20)
Capex to Assets	2.56***	2.74***	2.73***	2.54*	1.18	2.51***	3.10***
	(0.91)	(0.97)	(0.91)	(1.37)	(1.58)	(0.95)	(1.17)
Asset Growth	−0.11**	−0.15**	−0.08*	−0.24**	−0.21**	−0.10*	−0.11*
	(0.05)	(0.07)	(0.05)	(0.10)	(0.09)	(0.05)	(0.06)
Asset Turnover	0.92***	0.99***	0.77***	1.09***	0.96***	0.86***	0.70***
	(0.21)	(0.22)	(0.22)	(0.24)	(0.29)	(0.19)	(0.24)
Cash Holdings	−0.02	−0.01	−0.04	0.04	−0.02	−0.06	−0.06
	(0.06)	(0.07)	(0.05)	(0.08)	(0.06)	(0.06)	(0.06)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.97	0.97	0.97	0.97	0.97	0.97	0.96
# Obs.	1,241	1,241	1,241	1,238	1,119	1,241	1,241

This table presents difference-in-differences estimates of the impact of Washington State's Cap-and-Invest Program on seven corporate biodiversity footprint (CBF) outcomes estimated from Equation (1). Columns (1) through (7) correspond to total CBF, CBF-ex-GHG, CBF-GHG, CBF-Land Use, CBF-Resource Use, CBF-Air Pollution, and CBF-Water Pollution, respectively. The coefficients of interest are interaction terms between Treat and year indicators, with 2021 as the omitted base year. Standard errors (in parentheses) are clustered at the state level. Statistical significance is indicated by *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Please refer to Table A1 for variable definitions.

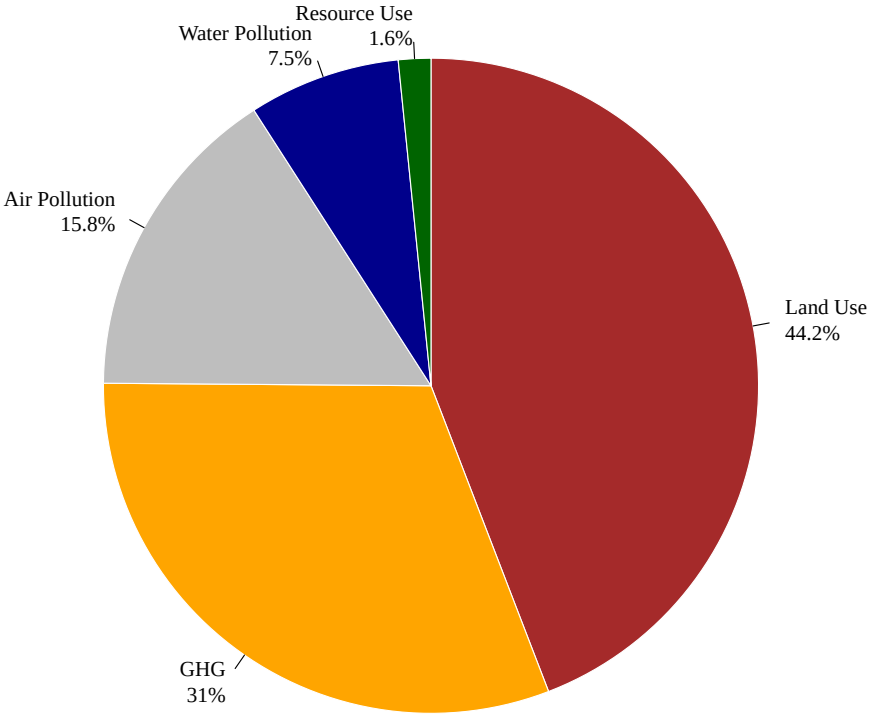
Table 4: Heterogeneous Treatment Effects Results

	Total		Ex-GHG		GHG		Land Use		Resource Use		Air Pol.		Water Pol.	
	(1) Pre	(2) Post	(3) Pre	(4) Post	(5) Pre	(6) Post	(7) Pre	(8) Post	(9) Pre	(10) Post	(11) Pre	(12) Post	(13) Pre	(14) Post
H1: Ln(CBF-GHG)	1.53	-0.02	3.27**	0.01	0.87	-0.03***	8.18***	0.03***	5.71***	-0.34***	9.17***	-0.01	4.23***	-0.01
	(0.2043)	(0.1073)	(0.0206)	(0.3900)	(0.4579)	(0.0002)	(0.0000)	(0.0079)	(0.0007)	(0.0000)	(0.0000)	(0.2043)	(0.0055)	(0.1408)
H2: Capex to Assets	1.85	-2.74**	4.83***	-1.20	0.81	-2.19***	16.27***	-1.38	28.88***	-15.01***	16.96***	8.86***	3.62**	4.49***
	(0.1362)	(0.0109)	(0.0024)	(0.2760)	(0.4869)	(0.0039)	(0.0000)	(0.2568)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0128)	(0.0000)
H3: Asset Growth	6.42***	0.56***	18.68***	0.71***	1.66	0.41**	118.33***	0.99***	31.35***	-0.33	2.30*	2.31***	3.21**	1.98***
	(0.0003)	(0.0042)	(0.0000)	(0.0002)	(0.1736)	(0.0315)	(0.0000)	(0.0000)	(0.0000)	(0.4901)	(0.0754)	(0.0000)	(0.0224)	(0.0000)
H4: Size	1.21	0.01	1.76	0.04***	1.54	-0.01	4.09***	0.08***	4.15***	-0.67***	1.03	0.03**	0.83	0.03**
	(0.3054)	(0.5116)	(0.1538)	(0.0074)	(0.2033)	(0.5017)	(0.0067)	(0.0000)	(0.0062)	(0.0000)	(0.3805)	(0.0360)	(0.4797)	(0.0475)

This table presents heterogeneous treatment effects across four firm-level characteristics: pre-policy GHG-related biodiversity exposure, capital investment intensity, asset growth, and firm size. The estimates are obtained from Equation (2) for seven corporate biodiversity footprint (CBF) outcomes. Column numbers index each Pre/Post pair by outcome. The *Pre* column reports the Wald/F statistic testing the joint significance of the pre-treatment triple-interaction coefficients (2018–2020). Rejection of the null indicates differential pre-treatment trends. The *Post* column reports the aggregated post-treatment interaction effect across 2022–2024, computed as the inverse-variance-weighted average of the corresponding yearly triple-interaction coefficients. Values in parentheses are p-values. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. P-values displayed as 0.0000 indicate values below 0.00005. See Table A1 for variable definitions.

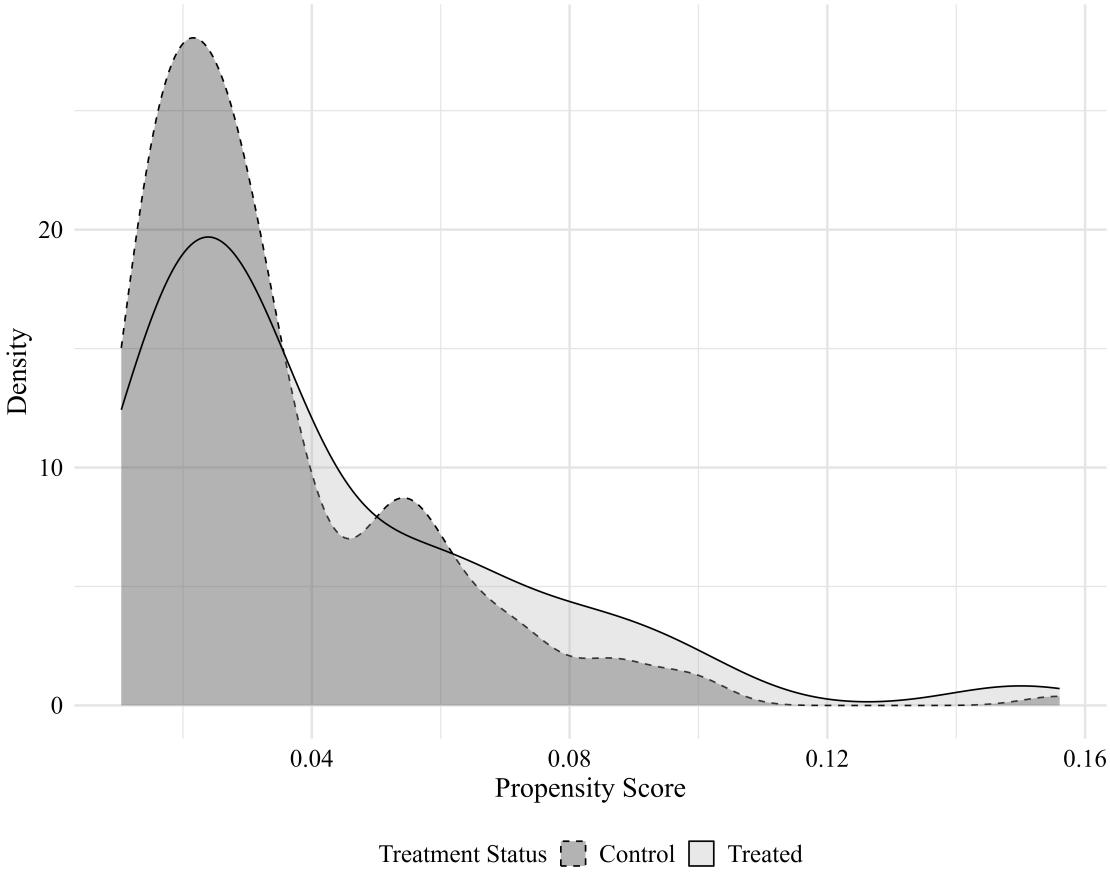
FIGURES

Figure 1: Source-based decomposition of the CBF—U.S. Firms



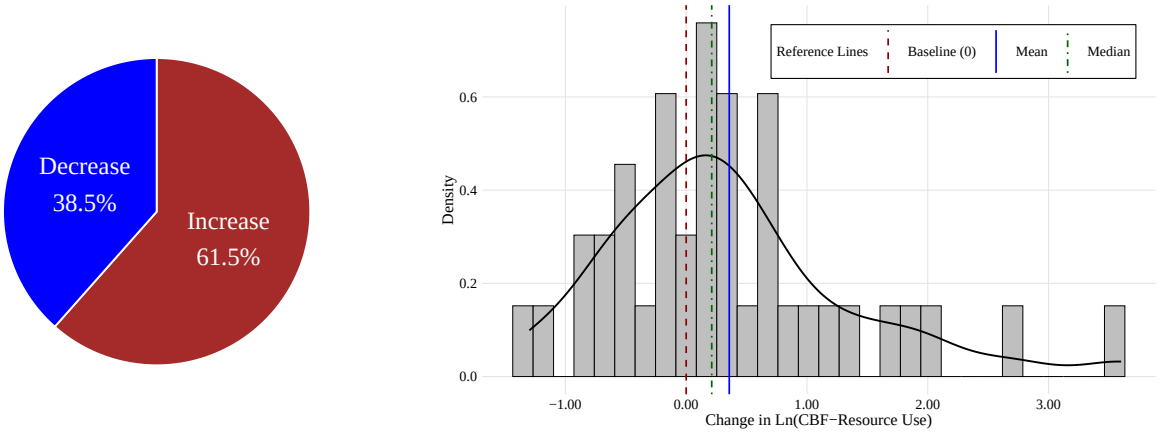
The pie chart shows the relative contribution of each environmental pressure category to the CBF of U.S. firms.

Figure 2: Propensity Score Matching Diagnostics



This figure displays the propensity score density distribution for treated (solid line, light gray) and control (dashed line, dark gray) firms.

Figure 3: Change in CBF-Resource Use Among Treated Firms Following the Policy

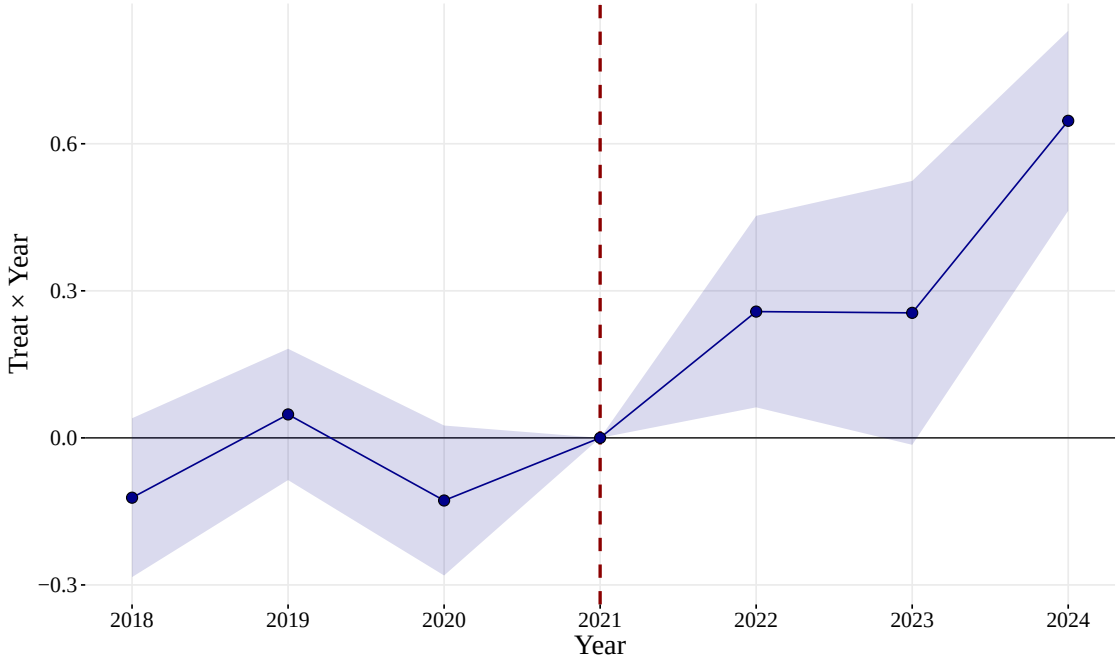


(a) Share of Firms

(b) Distribution of Firm-Level Changes

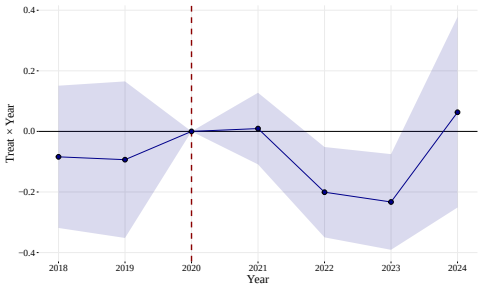
This figure illustrates changes in Ln(CBF-Resource Use) for treated firms following the introduction of Washington Cap-and-Invest Program. Panel (*a*) reports the share of treated firms that decrease or increase their CBF-Resource Use following the policy. Panel (*b*) presents the distribution of firm-level changes in Ln(CBF-Resource Use). The y-axis denotes density, and the x-axis shows the change in Ln(CBF-Resource Use). The vertical dashed line indicates zero (no change), the solid line indicates the mean change, and the dot-dashed line indicates the median change.

Figure 4: Dynamic Treatment Effects on CBF-GHG-Scope 3 Upstream

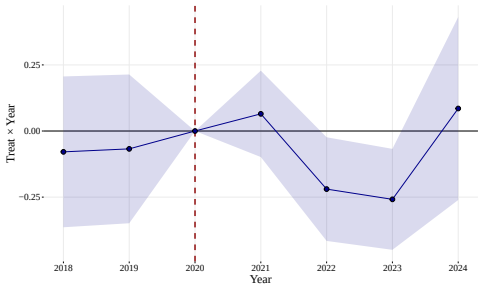


This figure displays the dynamic treatment effects for CBF-GHG-Scope 3 Upstream estimated using Equation (1). The coefficients correspond to interactions between the treatment indicator and year dummies, with 2021 serving as the omitted base year. The shaded areas represent 95% confidence intervals based on state-clustered robust standard errors. The vertical dashed line indicates the onset of the Washington State's Cap-and-Invest Program in 2021.

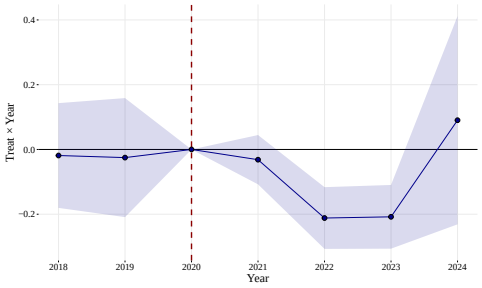
Figure 5: Dynamic Treatment Effects Under Placebo Treatment Timing



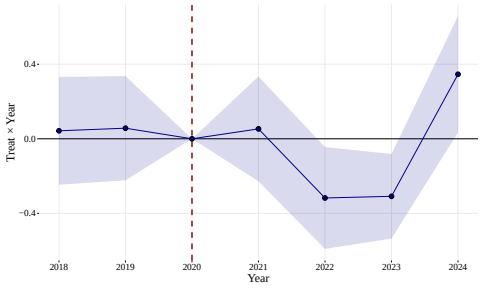
(a) CBF-Total



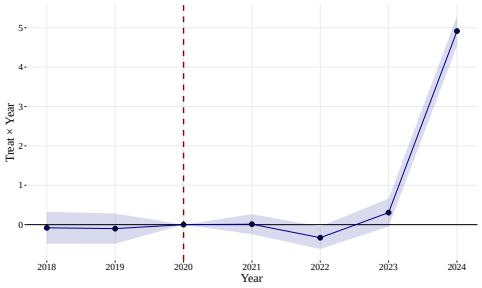
(b) CBF-ex-GHG



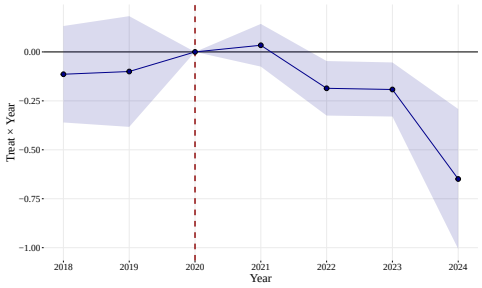
(c) CBF-GHG



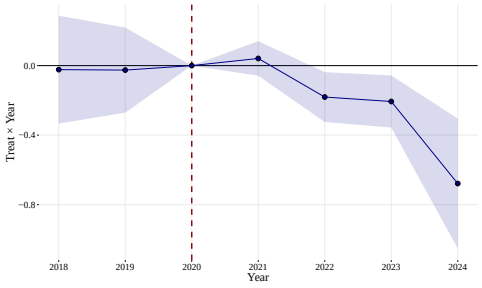
(d) CBF-Land Use



(e) CBF-Resource Use



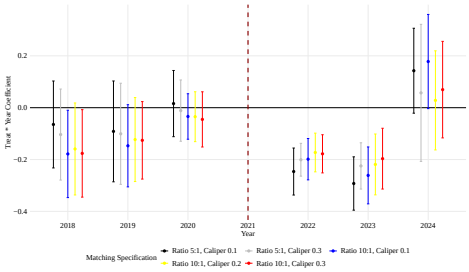
(f) CBF-Air Pollution



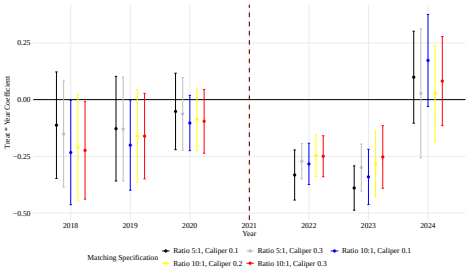
(g) CBF-Water Pollution

This figure reports dynamic treatment effects using a placebo treatment year set to 2020, which precedes the actual enactment of Washington State's Climate Commitment Act. The coefficients are estimated from Equation (1) and represent the interaction between the treatment indicator and year-specific dummies, with 2020 omitted as the base period. The vertical dashed line indicates the policy enactment year. The shaded areas represent 95% confidence intervals. The seven panels report results for (*a*) CBF-Total, (*b*) CBF-ex-GHG, (*c*) CBF-GHG, (*d*) CBF-Land Use, (*e*) CBF-Resource Use, (*f*) CBF-Air Pollution, and (*g*) CBF-Water Pollution.

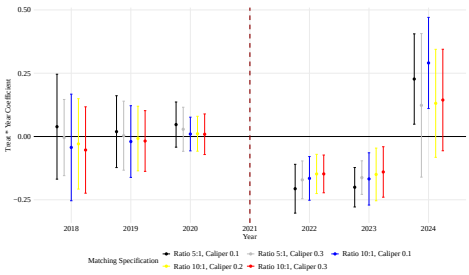
Figure 6: Dynamic Treatment Effects Under Alternative Matching Specifications



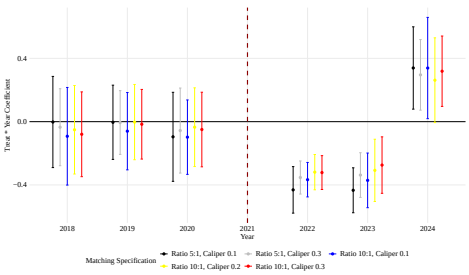
(a) CBF-Total



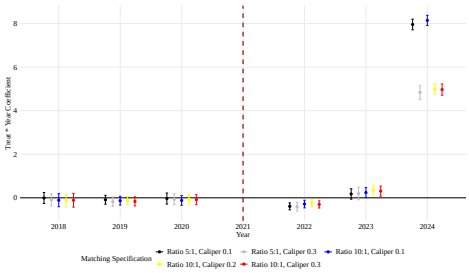
(b) CBF-ex-GHG



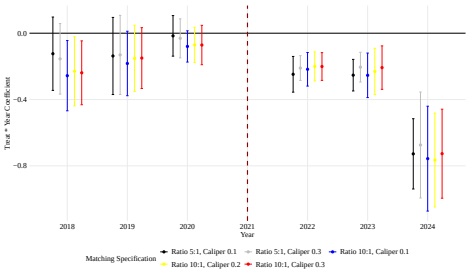
(c) CBF-GHG



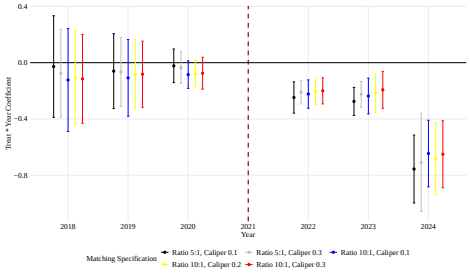
(d) CBF-Land Use



(e) CBF-Resource Use



(f) CBF-Air Pollution



(g) CBF-Water Pollution

This figure reports dynamic treatment effects using an alternative propensity score matching specifications with 5:1 nearest-neighbor matching with calipers of 0.1 and 0.3 (black and grey lines, respectively), as well as 10:1 nearest-neighbor matching with calipers of 0.1, 0.2, and 0.3 (blue, yellow, and red lines, respectively). The coefficients are estimated from Equation (1) and represent the interaction between the treatment indicator and year-specific dummies, with 2021 omitted as the base period. The vertical dashed line indicates the policy enactment year. For each specification, the point estimates and 95% confidence intervals are displayed. The seven panels report results for (*a*) CBF-Total, (*b*) CBF-ex-GHG, (*c*) CBF-GHG, (*d*) CBF-Land Use, (*e*) CBF-Resource Use, (*f*) CBF-Air Pollution, and (*g*) CBF-Water Pollution.

APPENDIX

Table A1: Variable Definitions

Variable	Definition
CBF-Total	Direct and indirect corporate biodiversity footprint covering all environmental impact channels, including emissions, land transformation, resource extraction, and pollution. It represents the overall firm-level biodiversity pressure across global supply chains and operational activities. Winsorized at the 1st and 99th percentiles.
CBF-ex-GHG	Corporate biodiversity footprint excluding the greenhouse gas emissions component, computed as CBF-Total net of CBF-GHG. It captures the aggregate biodiversity pressure attributable to non-climate channels, including land use, resource extraction, air pollution, and water pollution. Winsorized at the 1st and 99th percentiles.
CBF-GHG	Direct and indirect corporate biodiversity footprint attributable to greenhouse gas emissions, capturing climate-driven ecosystem degradation effects through radiative forcing and climate-related habitat disruption. Winsorized at the 1st and 99th percentiles.
CBF-GHG-Scope 3 Upstream	Component of the GHG-related biodiversity footprint attributable to upstream Scope 3 emissions embedded in purchased goods, services, and supply-chain inputs, as distinct from direct operational emissions (Scope 1) and purchased energy (Scope 2). This measure captures biodiversity pressure transmitted through a firm's upstream supply chain rather than its own production activities. Winsorized at the 1st and 99th percentiles.
CBF-Land Use	Direct and indirect corporate biodiversity footprint arising from land occupation, land transformation, and conversion of natural habitats into industrial, agricultural, or urban land. Winsorized at the 1st and 99th percentiles.
CBF-Resource Use	Direct and indirect corporate biodiversity footprint generated by extraction and consumption of natural resources (e.g., minerals, fossil fuels, biomass), including upstream supply-chain dependencies. Winsorized at the 1st and 99th percentiles.
CBF-Air Pollution	Direct and indirect corporate biodiversity footprint due to emissions of air pollutants (e.g., NOx, SOx, particulate matter), which affect ecosystem quality through atmospheric deposition and ecosystem toxicity. Winsorized at the 1st and 99th percentiles.
CBF-Water Pollution	Direct and indirect corporate biodiversity footprint associated with waterborne pollutants, including chemical discharges and nutrient loading affecting aquatic ecosystems. Winsorized at the 1st and 99th percentiles.
Treat	It is an indicator variable equal to one for firms in the treatment group and zero otherwise. The treatment group includes (*i*) firms identified in the CITSS registry as Covered Entities/Covered Sources that are subject to mandatory compliance obligations under Washington State's Cap-and-Invest Program, and (*ii*) publicly listed firms headquartered in Washington State that are not included in the CITSS registry.
Size	Natural logarithm of total book assets, capturing firm size and production scale. Winsorized at the 1st and 99th percentiles.
Book to Market	Book equity divided by market equity, commonly used as a valuation and distress-risk proxy. Winsorized at the 1st and 99th percentiles.
ROA	Return on assets, computed as net income divided by total assets, measuring accounting profitability. Winsorized at the 1st and 99th percentiles.
Leverage	Total debt divided by total assets, measuring the extent of asset financing through debt. Winsorized at the 1st and 99th percentiles.
Capex to Assets	Capital expenditures divided by total assets, capturing firm-level investment intensity. Winsorized at the 1st and 99th percentiles.
Asset Growth	Annual percentage change in total assets. Winsorized at the 1st and 99th percentiles.
Asset Turnover	Sales divided by total assets, to capture operational efficiency in generating revenues from the firm's asset base. Winsorized at the 1st and 99th percentiles.
Cash Holdings	Natural logarithm of the ratio of cash and short-term investments to total assets, to account for liquidity and internal financing capacity. Winsorized at the 1st and 99th percentiles.

This table defines all variables used in the empirical analysis.